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Supplementary appendix

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Supplementary appendix

Supplement to: Spatiotemporal patterns of Rift Valley fever virus in Africa: insights from genomic sequencing and phylodynamic modelling

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Appendix Text S1: RVFV data collection and curation

We used genomic data that was generated in our lab ([PRJNA1106221](#)) and combined with publicly available datasets retrieved from the National Center for Biotechnology (NCBI) [GenBank](#) nucleotide database¹. Our in-house genomic data were generated through amplicon multiplex polymerase chain reaction (PCR) and cell culture enrichment sequencing methods as described previously^{2,3}. The data consisted of retrospective human and livestock isolates as well as samples collected during syndromic surveillance conducted during outbreaks in Burundi (2018), Kenya (2007, 2018, 2019, 2021, and 2022) and Rwanda (2018 and 2022). Publicly available RVFV genomic sequences were obtained from NCBI using the keywords; “Rift Valley fever virus segment L AND 6404[SLEN]”, “Rift Valley fever virus segment M AND 3885[SLEN]” and “Rift Valley fever virus segment S AND 1520:1690[SLEN]” for complete large (n=215), medium (n=221) and small (n=234) segments respectively as of December 20, 2023. The search terms included the “SLEN” keyword to denote the sequence length for large and medium segments or sequence length range for the small segment. We searched in literature through the PUBMED database and obtained spatio-temporal information linked with the retrieved sequences. For downstream analyses, we retained sequences that only originated from the African continent and the Arabian Peninsula (Saudi Arabia). We applied several filtering criteria including: (i) including only sequences from Africa and the Arabian Peninsula, (ii) exclusion of laboratory strains (reassortants, antivirals or vaccine strains), (iii) exclusion of sequences without known geographical locations and (iv) deduplication in sequence composition using seqkit⁴. The global dataset consisted of large (n=236), medium (n=237) and small (n=247) following filtering steps as outlined in the methods section (main). The breakdown of these sequences is as follows: large segment (NCBI collection; n = 180, outbreak collection; n = 56), medium segment (NCBI collection; n = 184, outbreak collection; n = 53) and small segment (NCBI collection; n = 202, outbreak collection; n = 45).

Appendix Table S1: Number of RVFV genome sequences per host

Table showing the sequence collection consisting of datasets from the NCBI, and sequenced collection of outbreak samples generated from previous studies.

Host	Large segment	Medium segment	Small segment
Bat	1	2	1
Buffalo	5	4	6
Cow	74	73	1
Goat	2	2	75
Human	128	129	126
Mosquito	17	20	21
Sheep	9	7	16
Springbok (Antelope)			1

Total	236	237	247
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Appendix Table S3: Number of RVFV genome sequences per country

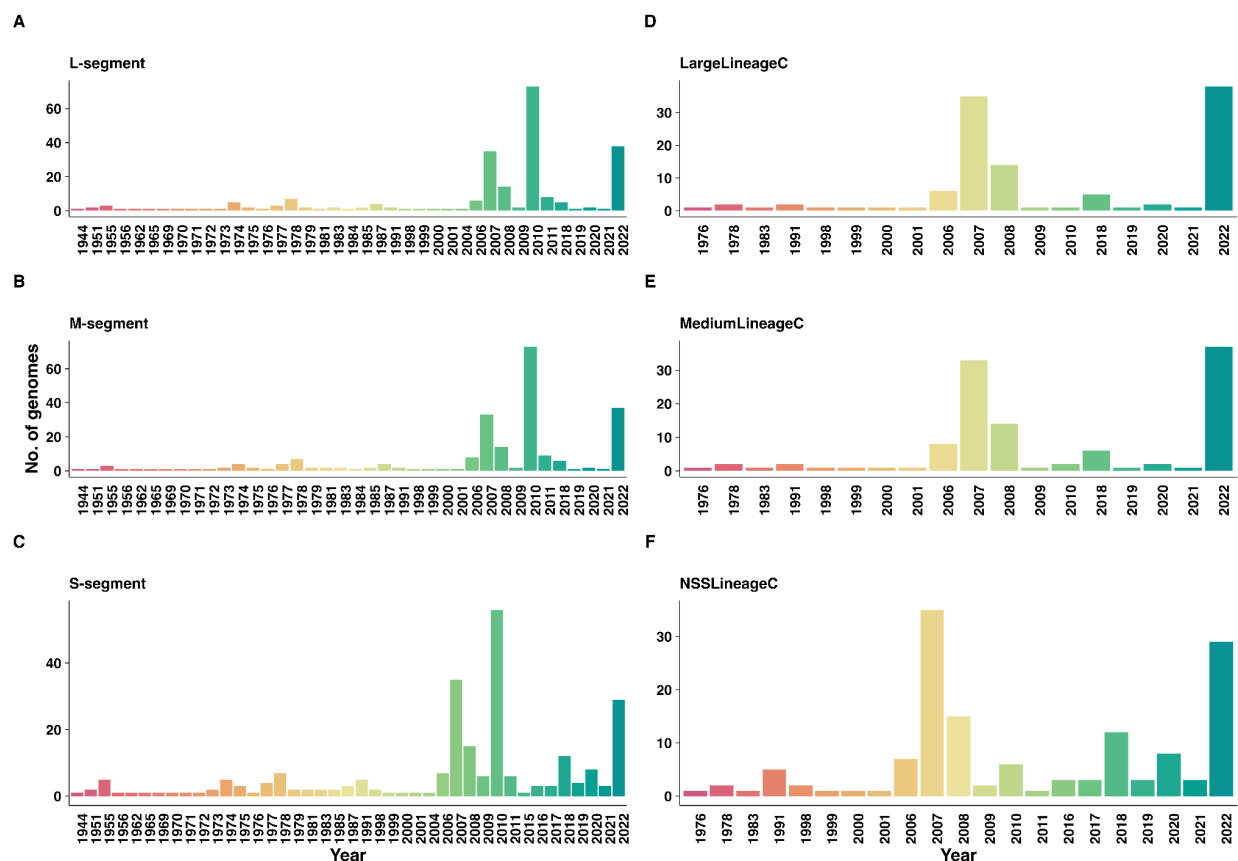
Rift Valley fever virus genome sequences retrieved from NCBI, and sequenced collection of outbreak samples generated from previous studies grouped by country of origin.

Country	Large segment	Medium segment	Small segment
Burkina Faso	1	1	1
Burundi	5	7	4
Central African Republic	5	5	5
Egypt	7	8	8
Guinea	1	2	1
Kenya	50	49	51
Madagascar	10	11	10
Mauritania	4	4	5
Mayotte	2	2	2
Namibia	2	2	3
Rwanda	35	32	27
Saudi Arabia	2	2	2
Senegal	1	1	1
South Africa	95	93	80
Sudan	1	2	10
Tanzania	4	4	3
Uganda	2	3	25
Zimbabwe	9	9	9
Total	236	237	247

Appendix Table S3: Final number of RVFV outbreak sequences used in this study grouped by country of origin

Rift Valley fever virus genome sequences recovered following sequencing of positive RVF samples.

Country	Large segment	Medium segment	Small segment
Burundi	5	7	4
Kenya	16	14	14
Rwanda	35	32	27
Total	56	53	45



Appendix Figure S1. Number of RVFV genomes per year. (A-C) Bar plot showing the number of all RVFV genomes collected between 1944 and 2022, (D-F) Bar plot showing the number of all RVFV lineage C genomes collected between 1976 and 2022.

Appendix Text S2: Multiple sequence alignment and phylogenetic inference

For each genomic segment dataset, multiple sequence alignment was performed using MAFFT (version 7.475)⁵ applying FFT-NS-2 strategy with progressive guide trees. The alignments were manually inspected in AliView (version 1.28)⁶ and primer-binding sites trimmed to obtain correct codon alignments. The resulting gene-specific datasets consisted of RNA-dependent RNA polymerase (large), glycoprotein (medium), non-structural and nucleocapsid sequences (small). Optimal evolutionary models were identified using jModelTest (version 0.1.7)⁷ and implemented in phylogenetic inference methods of maximum likelihood and Bayesian phylogenetics approaches. A general time-reversible (GTR)⁸ with a discrete gamma distributed rate heterogeneity model with invariant sites (GTR+I+G4) was the best-fitting model for all the datasets. Maximum likelihood tree reconstruction was performed using IQ-TREE (version 2.0.3)⁹ with fast (-bb) bootstrapping of 1,000 replicates. To estimate the root-to-tip distances

regressed against time, we used TempEst (version 1.5.3)¹⁰ by applying the best-fitting root with a residual mean squared function from the resulting maximum likelihood trees. This inspection revealed sufficiently strong temporal signals for molecular clock analysis using TreeTime (version 0.93)¹¹ and BEAST (version 1.10.4)¹².

Appendix Text S3: Bayesian phylogeographic inference in continuous space

To deduce the dispersal history of viruses from sequence or any other data, it is essential to model how the pathogens spread through space^{13,14}. To characterise the dispersal history and dynamics of RVFV in a continuous diffusion approach, a relaxed random walk (RRW) model was applied^{15,16}. Its implementation allows for inference of spatially and temporally referenced phylogenies while accommodating variation in dispersal velocity among branches¹⁷. To model among-branch heterogeneity in diffusion velocity, a Cauchy probability distribution was used in the phylogeographic inference. The bivariate traits were represented as latitude and longitude with addition of a random jitter to tips at a window of 0.05. The settings of this continuous phylogeographic analysis also included a GTR+I+G4 substitution model while accommodating for branch-specific evolutionary rate using the uncorrelated log-normal relaxed clock and a skygrid prior as the coalescent population growth model. BEAST was run with 1.5×10^9 Markov chain Monte-Carlo (MCMC) steps, sampling every 150 000th step for the large segment, 1.2×10^9 MCMC steps, sampling every 120 000th step for the medium and 6.0×10^8 MCMC steps, sampling every 60 000th step for the small segment. Broad-platform Evolutionary Analysis General Likelihood Evaluator (BEAGLE) library was incorporated in the analyses to speed-up the likelihood approximation process¹⁸. The convergence and mixing properties of the outputs were inspected to ensure effective sample size values linked with estimated parameters were all >200 using Tracer (version 1.7.2)¹⁹. TreeAnnotator¹² was used to retrieve and annotate the maximum clade credibility (MCC) tree.

Appendix Text S4: Landscape phylogeographic analyses

Landscape phylogeographic frameworks can exploit spatially explicit phylogeographic reconstructions to investigate the impact of environmental factors on the dispersal of viral lineages²⁰. To summarise RVFV diffusion over time and space, 300 post-burn-in trees sampled at regular intervals were extracted from the posterior distribution of each phylogeographic reconstruction. The R package “seraphim”²¹ was used to extract spatiotemporal information embedded in posterior trees and to estimate dissemination statistics such as the weighted diffusion coefficients²² and the evolution of the maximal wavefront distance from the epidemic origin²¹. We tested the impact of several environmental factors on the diffusion velocity of RVFV lineages. These environmental factors included elevation, the normalised difference vegetative index (NDVI), precipitation, temperature, croplands, pastures, urban areas, forest cover, cattle density, soil available water storage capacity, topsoil bulk density, soil type and

grasslands. Each environmental raster was once tested as a potential resistance factor (impeding the movement of the virus) and once tested as a conductance factor (facilitating the movement of the virus). For each environmental factor, we also transformed the original raster cell values using the following formula: $v_t = 1 + k \cdot (v_0/v_{max})$, where v_t is the transformed cell value, v_0 the original cell value, and v_{max} the maximum cell value in the raster. The rescaling parameter k offers the possibility to test different strengths of raster cell resistance or conductance in relation to the resistance/conductance of a cell with a minimum value set to “1”. For all the environmental factors, three different k parameter values (10, 100, and 1000) were tested.

The investigation of the impact of environmental factors on RVFV lineage diffusion velocity consisted in four successive steps²³: for a given environmental raster to test, (i) an environmental distance was computed for each phylogenetic branch using the Circuitscape algorithm that uses circuit theory to allow integrating uncertainty in the route taken during the dispersal of viral lineages²³ (ii) The correlation between the phylogenetic branch duration and environmental distance associated with each phylogenetic branch was then estimated. For this purpose, we estimated a Q statistic for each sampled tree; a statistic being defined as the difference between two coefficients of determination ($Q = R^2_{env} - R^2_{null}$): (i) the coefficient of determination R^2_{env} obtained from the linear regression between four times the phylogenetic branch durations t and the squared environmental distances d^2_{env} computed on the environmental raster to test ($4t \sim d^2_{env}$), and (ii) the coefficient of determination R^2_{null} obtained from the linear regression between four times the phylogenetic branch durations t and the squared environmental distances d^2_{env} computed on the “null” raster ($4t \sim d^2_{null}$), i.e. a raster with the value of all accessible cells set to “1”. An environmental raster was considered as potentially explanatory only if the distribution of regression coefficients and associated distribution of Q statistic values were positive²⁴. (iii) A null dispersal model is obtained by randomising the phylogenetic branch positions within the study area (by maintaining the tree topology and the position inferred for the root of each trees, as well as by preventing that tree nodes fall into non-accessible areas like, e.g., sea areas). We conducted one randomisation per sampled tree and therefore obtained two distributions of Q values: one distribution from the inferred and the other from the randomised trees. (iv) We estimated the statistical support associated with each positive distribution of Q values through the computation of a Bayes Factor (BF) support: $BF = [p_e/(1-p_e)]/[0.5/(1-0.5)]$, where p_e is the posterior probability that $Q_{inferred} > Q_{randomised}$.

Finally, we also tested the association between the RVFV lineage dispersal locations and the environmental factors listed above. This second analytical approach began by the computation of the E statistic that calculates the mean environmental values at tree node positions²⁰. The E statistic can be used to investigate the tendency of inferred lineages to preferentially circulate or

to avoid circulating in specific environmental conditions²⁰. The statistical support for the estimated *E* statistics was again assessed using the above-described randomisation procedure.

Appendix Table S4: RVFV large segment genome accessions and metadata used in the phylogenetic and phylogeographic analysis

Table showing the sequence genomes for the large segment of RVFV providing information on the lineage, host, country, location and year of collection. The source column indicates publicly accessed sequences (NCBI) and those sequenced as part of the RVF studies in East Africa (Kenya, Rwanda and Burundi) from outbreak samples and samples archived from sporadic RVF outbreaks.

Accession	Lineage	Host	Country	Location	Year	Source
DQ375395	N	Human	Mauritania	Kiffa	1987	NCBI
DQ375396	N	Human	Mauritania	Selibaby	1987	NCBI
DQ375397	N	Human	Mauritania	Ayoum el Atrous	1987	NCBI
DQ375398	N	Human	Mauritania	Keur Macene	1987	NCBI
DQ375399	N	Mosquito	Burkina Faso	Fada-Ngourma	1983	NCBI
DQ375400	C	Human	Kenya	Garissa	1998	NCBI
DQ375401	C	Human	Saudi Arabia	Jizan	2000	NCBI
DQ375402	C	Mosquito	Kenya	Ruiru	1983	NCBI
DQ375405	A	Mosquito	Egypt	Sharqiya	1977	NCBI
DQ375407	A	Mosquito	Egypt	Sharqiya	1977	NCBI
DQ375409	A	Mosquito	Egypt	Sharqiya	1978	NCBI
DQ375410	A	Sheep	Egypt	Gharbiya	1979	NCBI
DQ375411	A	Human	Egypt	Gharbiya	1978	NCBI
DQ375412	A	Cow	Egypt	Asyut	1978	NCBI
DQ375413	A	Cow	Zimbabwe	Beatrice	1974	NCBI
DQ375414	A	Cow	Madagascar	Moramanga	1979	NCBI
DQ375415	E	Human	Central African Republic	Ombella Mpoko	1974	NCBI
DQ375418	E	Cow	Zimbabwe	Salisbury	1978	NCBI
DQ375419	G	Human	Central African Republic	Bangui	1969	NCBI
DQ375421	G	Bat	Guinea	Macenta	1984	NCBI
DQ375422	E	Human	Central African Republic	Mbaki	1985	NCBI
DQ375423	E	Human	Central African Republic	Bangui	1985	NCBI
DQ375424	E	Cow	Zimbabwe	Sinoia	1978	NCBI
DQ375425	D	Human	Central African Republic	Ombella Mpoko	1973	NCBI
DQ375426	L	Cow	Zimbabwe	Salisbury	1970	NCBI
DQ375427	L	Cow	Kenya	Kiambu	1965	NCBI
DQ375428	L	Human	South Africa	Randfontein	1975	NCBI
DQ375429	K	Human	Uganda	Entebbe	1944	NCBI
DQ375431	K	Sheep	Kenya	Kabete	1951	NCBI
DQ375432	K	Cow	Zimbabwe	Salisbury	1974	NCBI
DQ375433	O	Sheep	South Africa	Boshof	1951	NCBI
DQ375434	J	Cow	Zimbabwe	Sinoia	1974	NCBI
EU574004	C	Cow	Kenya	Kiambu	2007	NCBI
EU574005	C	Cow	Kenya	Nairobi	2007	NCBI
EU574006	C	Sheep	Kenya	Baringo	2007	NCBI
EU574007	C	Cow	Kenya	Embu	2007	NCBI
EU574008	C	Buffalo	Kenya	Garissa	2007	NCBI
EU574009	C	Cow	Kenya	Thika	2007	NCBI
EU574010	C	Cow	Kenya	Nakuru	2007	NCBI
EU574011	C	Cow	Kenya	Meru Central	2007	NCBI
EU574013	C	Cow	Kenya	Nairobi	2007	NCBI
EU574014	C	Goat	Kenya	Garissa	2006	NCBI
EU574015	C	Goat	Kenya	Garissa	2006	NCBI

EU574017	C	Cow	Kenya	Muranga	2007	NCBI
EU574019	C	Cow	Kenya	Meru South	2007	NCBI
EU574020	C	Cow	Kenya	Thika	2007	NCBI
EU574025	C	Cow	Kenya	Maragua	2007	NCBI
EU574028	C	Cow	Kenya	Maragua	2007	NCBI
EU574029	C	Cow	Kenya	Garissa	2007	NCBI
HE687304	C	Human	Mayotte	Bandraboua	2008	NCBI
HE687305	C	Human	Mayotte	Sada	2008	NCBI
HM586953	C	Human	Kenya	Garissa	2006	NCBI
HM586954	C	Human	Kenya	Garissa	2006	NCBI
HM586955	C	Human	Kenya	Kilifi	2007	NCBI
HM586956	C	Human	Kenya	Malindi	2007	NCBI
HM586957	C	Human	Kenya	Baringo	2007	NCBI
HM586958	C	Human	Kenya	Kiserian	2007	NCBI
HM586959	C	Human	Tanzania	Tanga	2007	NCBI
HM586960	C	Human	Tanzania	Dodoma	2007	NCBI
HM586961	C	Mosquito	Kenya	Garissa	2006	NCBI
HM586962	C	Mosquito	Kenya	Kilifi	2007	NCBI
HM586963	C	Mosquito	Kenya	Kiserian	2007	NCBI
JF311368	C	Human	Madagascar	Taolagnaro	2008	NCBI
JF311369	C	Human	Madagascar	Anjozorobe	2008	NCBI
JF311370	C	Human	Madagascar	Ankazobe	2008	NCBI
JF311371	C	Human	Madagascar	Manjakandriana	2008	NCBI
JF311372	C	Human	Madagascar	Antananarivo	1991	NCBI
JF311373	C	Cow	Madagascar	Antananarivo	1991	NCBI
JF311374	C	Cow	Madagascar	Miarinarivo	2008	NCBI
JF311375	C	Cow	Madagascar	Antsirabe	2008	NCBI
JF311376	C	Cow	Madagascar	Manjakandriana	2008	NCBI
JF326186	C	Mosquito	Kenya	Garissa	2007	NCBI
JF326189	C	Human	Tanzania	Dodoma	2007	NCBI
JF326190	C	Human	Tanzania	Dodoma	2007	NCBI
JF784386	L	Sheep	South Africa	Free State	1974	NCBI
JQ820485	C	Human	Sudan	Gezira	2010	NCBI
KX096938	C	Mosquito	Kenya	Garissa	2006	NCBI
KX096941	C	Mosquito	Saudi Arabia	Aseer	2001	NCBI
KX944851	H	Cow	South Africa	Groenhof	2009	NCBI
KX944862	H	Sheep	South Africa	Middelburg	2010	NCBI
KX944863	C	Buffalo	South Africa	Hoedspruit	2008	NCBI
KX944864	C	Buffalo	South Africa	Mbombela	2008	NCBI
KX944865	C	Cow	South Africa	Bela-Bela	2008	NCBI
KY126665	H	Human	South Africa	Northern cape	2010	NCBI
KY126666	H	Human	South Africa	Free state	2010	NCBI
KY126667	H	Human	South Africa	Northern cape	2010	NCBI
KY126669	H	Human	South Africa	Free state	2010	NCBI
KY126670	H	Human	South Africa	Free state	2010	NCBI
KY126671	H	Human	South Africa	Free state	2010	NCBI
KY126672	H	Human	South Africa	Free state	2010	NCBI
KY126673	H	Human	South Africa	Free state	2010	NCBI
KY126674	H	Human	South Africa	Free state	2010	NCBI
KY126675	H	Human	South Africa	Free state	2010	NCBI
KY126676	H	Human	South Africa	Eastern cape	2011	NCBI
KY126677	H	Human	South Africa	Eastern cape	2011	NCBI
KY126678	H	Human	Namibia	Caprivi	2004	NCBI
KY126680	C	Buffalo	South Africa	Mpumalanga	2008	NCBI
KY126681	C	Cow	South Africa	Bela-Bela	2008	NCBI
KY126683	C	Cow	South Africa	Northern cape	2009	NCBI
KY126685	I	Mosquito	South Africa	Free State	1955	NCBI
KY126686	I	Mosquito	South Africa	Free State	1956	NCBI
KY126687	L	Mosquito	South Africa	Standerton	1971	NCBI
KY126688	F	Mosquito	South Africa	Natal	1981	NCBI
KY126689	C	Buffalo	South Africa	Limpopo	1999	NCBI
MG659819	K	Mosquito	South Africa	Free State	1955	NCBI

[illegible]

MG659891	H	Human	South Africa	Eastern cape	2011	NCBI
MG659892	H	Human	South Africa	Eastern cape	2011	NCBI
MG659894	H	Human	South Africa	Eastern cape	2011	NCBI
MG659895	H	Human	South Africa	Eastern cape	2011	NCBI
MG659897	H	Human	Namibia	Caprivi	2011	NCBI
MG659898	K	Cow	Kenya	Makueni	1962	NCBI
MG659899	B	Sheep	Kenya	Mombasa	1972	NCBI
MG659900	C	Human	Zimbabwe	Gatooma	1976	NCBI
MG659901	C	Sheep	Zimbabwe	Sinoia	1978	NCBI
MG659902	C	Cow	Zimbabwe	Salisbury	1978	NCBI
MG659904	N	Human	Senegal	Kedougou	1975	NCBI
NC 01439	A	Human	Egypt	Sharqiya	1977	NCBI
7						
OQ440142	K	Mosquito	Uganda	Lunyo	1955	NCBI
OR780625	C	Cow	Burundi	Ruyaga	2022	Study
OR780626	C	Cow	Burundi	Ngozi	2022	Study
OR780627	C	Cow	Burundi	Kobero	2022	Study
OR780628	C	Cow	Burundi	Kobero	2022	Study
OR780629	C	Cow	Burundi	Kobero	2022	Study
PP746304	C	Cow	Rwanda	Nyarugenge	2022	Study
PP746305	C	Cow	Rwanda	Gicumbi	2022	Study
PP746306	C	Cow	Rwanda	Gicumbi	2022	Study
PP746307	C	Cow	Rwanda	Gicumbi	2022	Study
PP746308	C	Cow	Rwanda	Gicumbi	2022	Study
PP746309	C	Cow	Rwanda	Kamonyi	2022	Study
PP746310	C	Cow	Rwanda	Nyaruguru	2022	Study
PP746311	C	Cow	Rwanda	Nyanza	2022	Study
PP746313	C	Cow	Rwanda	Kamonyi	2022	Study
PP746314	C	Cow	Rwanda	Nyarugenge	2022	Study
PP746315	C	Cow	Rwanda	Muhanga	2022	Study
PP746316	C	Cow	Rwanda	Muhanga	2022	Study
PP746317	C	Cow	Rwanda	Rusizi	2022	Study
PP746318	C	Cow	Rwanda	Nyamasheke	2022	Study
PP746321	C	Cow	Rwanda	Nyarugenge	2022	Study
PP746322	C	Cow	Rwanda	Nyarugenge	2022	Study
PP746323	C	Cow	Rwanda	Nyarugenge	2022	Study
PP746324	C	Cow	Rwanda	Muhanga	2022	Study
PP746325	C	Cow	Rwanda	Muhanga	2022	Study
PP746326	C	Cow	Rwanda	Kamonyi	2022	Study
PP746327	C	Cow	Rwanda	Nyarugenge	2022	Study
PP746328	C	Cow	Rwanda	Gicumbi	2022	Study
PP746329	C	Cow	Rwanda	Kamonyi	2022	Study
PP746330	C	Cow	Rwanda	Nyarugenge	2022	Study
PP746333	C	Cow	Rwanda	Gicumbi	2022	Study
PP746334	C	Cow	Rwanda	Kamonyi	2022	Study
PP746335	C	Sheep	Rwanda	Muhanga	2022	Study
PP746336	C	Cow	Rwanda	Muhanga	2022	Study
PP746337	C	Cow	Rwanda	Nyarugenge	2022	Study
PP746338	C	Cow	Rwanda	Nyarugenge	2022	Study
PP746339	C	Cow	Rwanda	Nyarugenge	2022	Study
PP746340	C	Cow	Rwanda	Nyarugenge	2022	Study
PP746341	C	Cow	Rwanda	Nyarugenge	2022	Study
PP746415	C	Human	Kenya	Wajir	2018	Study
PP746424	C	Cow	Kenya	Marsabit	2018	Study
PP746425	C	Human	Kenya	Baringo	2007	Study
PP746426	C	Human	Kenya	Baringo	2007	Study
PP746427	C	Human	Kenya	Kilifi	2007	Study
PP746436	C	Human	Kenya	Baringo	2007	Study
PP746437	C	Human	Kenya	Kilifi	2007	Study
PP746438	C	Human	Kenya	Baringo	2007	Study
PP746445	C	Human	Kenya	Isiolo	2020	Study
PP746447	C	Human	Kenya	Kilifi	2007	Study

PP746448	C	Human	Kenya	Nairobi	2007	Study
PP746449	C	Human	Kenya	Baringo	2007	Study
PP746450	C	Human	Kenya	Nyandarua	2019	Study
PP746452	C	Human	Kenya	Isiolo	2020	Study
PP747229	C	Cow	Kenya	Wajir	2018	Study
PP747230	C	Cow	Kenya	Kiambu	2021	Study
PP747231	C	Cow	Rwanda	Ngoma	2018	Study
PP747233	C	Cow	Rwanda	Rulindo	2018	Study

Appendix Table S5: RVFV medium genome accessions and metadata used in the phylogenetic and phylogeographic analysis

Table showing the sequence genomes for the large segment of RVFV providing information on the lineage, host, country, location and year of collection. The source column indicates publicly accessed sequences (NCBI) and those sequenced as part of the RVF studies in East Africa (Kenya, Rwanda and Burundi) from outbreak samples and samples archived from sporadic RVF outbreaks.

Accession	Lineage	Host	Country	Location	Year	Source
DQ380183	N	Human	Mauritania	Ayoune el Atrous	1987	NCBI
DQ380184	N	Human	Mauritania	Selibaby	1987	NCBI
DQ380185	N	Human	Mauritania	Kiffa	1987	NCBI
DQ380186	N	Human	Mauritania	Keur Macene	1987	NCBI
DQ380187	N	Mosquito	Burkina Faso	Fada-Ngourma	1983	NCBI
DQ380188	L	Cow	Zimbabwe	Salisbury	1970	NCBI
DQ380189	L	Human	South Africa	Randfontein	1975	NCBI
DQ380190	L	Cow	Kenya	Kiambu	1965	NCBI
DQ380191	K	Human	Uganda	Entebbe	1944	NCBI
DQ380194	K	Cow	Zimbabwe	Salisbury	1974	NCBI
DQ380195	O	Sheep	South Africa	Boshof	1951	NCBI
DQ380196	C	Human	Kenya	Garissa	1998	NCBI
DQ380197	C	Human	Saudi Arabia	Jizan	2000	NCBI
DQ380198	C	Mosquito	Kenya	Ruiru	1983	NCBI
DQ380199	A	Mosquito	Egypt	Sharqiya	1977	NCBI
DQ380200	A	Human	Egypt	Sharqiya	1977	NCBI
DQ380201	A	Mosquito	Egypt	Sharqiya	1977	NCBI
DQ380203	A	Human	Egypt	Gharbiya	1978	NCBI
DQ380204	A	Mosquito	Egypt	Sharqiya	1978	NCBI
DQ380205	A	Sheep	Egypt	Gharbiya	1979	NCBI
DQ380207	A	Cow	Egypt	Asyut	1978	NCBI
DQ380209	A	Cow	Zimbabwe	Beatrice	1974	NCBI
DQ380210	A	Cow	Madagascar	Moramanga	1979	NCBI
DQ380211	E	Human	Central African Republic	Ombella Mpoko	1973	NCBI
DQ380214	E	Cow	Zimbabwe	Salisbury	1978	NCBI
DQ380215	G	Bat	Guinea	Macenta	1981	NCBI
DQ380216	G	Bat	Guinea	Macenta	1984	NCBI
DQ380217	G	Human	Central African Republic	Bangui	1969	NCBI
DQ380218	G	Human	Central African Republic	Mbaiki	1985	NCBI
DQ380219	G	Human	Central African Republic	Bangui	1985	NCBI
DQ380220	G	Cow	Zimbabwe	Sinoia	1978	NCBI
DQ380221	D	Human	Central African Republic	Ombella Mpoko	1973	NCBI
DQ380222	J	Cow	Zimbabwe	Sinoia	1974	NCBI
EF460404	C	Mosquito	Kenya	Garissa	2007	NCBI
EF467177	C	Mosquito	Kenya	Kilifi	2007	NCBI
EF467178	C	Mosquito	Kenya	Garissa	2006	NCBI
EU574031	C	Cow	Kenya	Kiambu	2007	NCBI
EU574032	C	Cow	Kenya	Nairobi	2007	NCBI
EU574034	C	Cow	Kenya	Embu	2007	NCBI
EU574035	C	Cow	Kenya	Thika	2007	NCBI
EU574036	C	Cow	Kenya	Nakuru	2007	NCBI

EU574037	C	Cow	Kenya	Meru Central	2007	NCBI
EU574039	C	Cow	Kenya	Nairobi	2007	NCBI
EU574040	C	Goat	Kenya	Garissa	2006	NCBI
EU574041	C	Goat	Kenya	Garissa	2006	NCBI
EU574042	C	Cow	Kenya	Meru South	2007	NCBI
EU574043	C	Cow	Kenya	Mombassa	2007	NCBI
EU574044	C	Cow	Kenya	Muranga	2007	NCBI
EU574046	C	Cow	Kenya	Meru South	2007	NCBI
EU574047	C	Cow	Kenya	Thika	2007	NCBI
EU574055	C	Cow	Kenya	Garissa	2007	NCBI
EU574056	C	Cow	Kenya	Maragua	2007	NCBI
HE687303	C	Human	Mayotte	Bandraboua	2008	NCBI
HE687306	C	Human	Mayotte	Sada	2008	NCBI
HM586964	C	Human	Kenya	Garissa	2006	NCBI
HM586965	C	Human	Kenya	Garissa	2006	NCBI
HM586966	C	Human	Kenya	Kilifi	2007	NCBI
HM586967	C	Human	Kenya	Malindi	2007	NCBI
HM586968	C	Human	Kenya	Baringo	2007	NCBI
HM586969	C	Human	Kenya	Kiserian	2007	NCBI
HM586970	C	Human	Tanzania	Tanga	2007	NCBI
HM586971	C	Human	Tanzania	Dodoma	2007	NCBI
HM586972	C	Mosquito	Kenya	Garissa	2006	NCBI
HM586973	C	Mosquito	Kenya	Kilifi	2007	NCBI
HQ009512	C	Cow	Madagascar	Antsirabe	2008	NCBI
JF311377	C	Human	Madagascar	Taolagnaro	2008	NCBI
JF311378	C	Human	Madagascar	Anjozorobe	2008	NCBI
JF311379	C	Human	Madagascar	Ankazobe	2008	NCBI
JF311380	C	Human	Madagascar	Manjakandriana	2008	NCBI
JF311381	C	Human	Madagascar	Antananarivo	1991	NCBI
JF311382	C	Cow	Madagascar	Antananarivo	1991	NCBI
JF311383	C	Cow	Madagascar	Miarinarivo	2008	NCBI
JF311384	C	Cow	Madagascar	Antsirabe	2008	NCBI
JF311385	C	Cow	Madagascar	Manjakandriana	2008	NCBI
JF326191	C	Mosquito	Kenya	Garissa	2007	NCBI
JF326192	C	Mosquito	Kenya	Garissa	2006	NCBI
JF326193	C	Mosquito	Kenya	Garissa	2006	NCBI
JF326194	C	Human	Tanzania	Dodoma	2007	NCBI
JF326195	C	Human	Tanzania	Dodoma	2007	NCBI
JF784387	L	Sheep	South Africa	Free State	1974	NCBI
JQ820487	C	Human	Sudan	Gezira	2010	NCBI
JQ820488	C	Human	Sudan	Gezira	2010	NCBI
KX096942	C	Mosquito	Saudi Arabia	Aseer	2001	NCBI
KX944840	C	Buffalo	South Africa	Hoedspruit	2008	NCBI
KX944841	C	Buffalo	South Africa	Mbombela	2008	NCBI
KX944842	C	Cow	South Africa	Bela-Bela	2008	NCBI
KY126690	H	Human	South Africa	Northern cape	2010	NCBI
KY126691	H	Human	South Africa	Free state	2010	NCBI
KY126692	H	Human	South Africa	Northern cape	2010	NCBI
KY126694	H	Human	South Africa	Free state	2010	NCBI
KY126695	H	Human	South Africa	Free state	2010	NCBI
KY126696	H	Human	South Africa	Free state	2010	NCBI
KY126697	H	Human	South Africa	Free state	2010	NCBI
KY126698	H	Human	South Africa	Free state	2010	NCBI
KY126699	H	Human	South Africa	Free state	2010	NCBI
KY126700	H	Human	South Africa	Free state	2010	NCBI
KY126701	H	Human	South Africa	Eastern cape	2011	NCBI
KY126702	H	Human	South Africa	Eastern cape	2011	NCBI
KY126705	C	Buffalo	South Africa	Mpumalanga	2008	NCBI
KY126708	C	Cow	South Africa	Northern cape	2009	NCBI
KY126709	H	Human	South Africa	Eastern cape	2009	NCBI
KY126710	I	Mosquito	South Africa	Free State	1955	NCBI
KY126711	I	Mosquito	South Africa	Free State	1956	NCBI

KY126712	L	Mosquito	South Africa	Standerton	1971	NCBI
KY126713	F	Mosquito	South Africa	Natal	1981	NCBI
KY126714	C	Buffalo	South Africa	Limpopo	1999	NCBI
MG659733	M	Mosquito	South Africa	Free State	1955	NCBI
MG659734	H	Human	South Africa	Northern cape	2010	NCBI
MG659735	H	Human	South Africa	Free state	2010	NCBI
MG659736	H	Human	South Africa	Northern cape	2010	NCBI
MG659737	H	Human	South Africa	Free state	2010	NCBI
MG659738	H	Human	South Africa	Free state	2010	NCBI
MG659739	H	Human	South Africa	Free state	2010	NCBI
MG659740	H	Human	South Africa	Free state	2010	NCBI
MG659741	H	Human	South Africa	Northern cape	2010	NCBI
MG659742	H	Human	South Africa	Free state	2010	NCBI
MG659743	H	Human	South Africa	Free state	2010	NCBI
MG659744	H	Human	South Africa	Eastern cape	2010	NCBI
MG659745	H	Human	South Africa	Eastern cape	2010	NCBI
MG659747	H	Human	South Africa	Free state	2010	NCBI
MG659748	H	Human	South Africa	Free state	2010	NCBI
MG659749	H	Human	South Africa	Free state	2010	NCBI
MG659750	H	Human	South Africa	Eastern cape	2010	NCBI
MG659752	H	Human	South Africa	Free state	2010	NCBI
MG659754	H	Human	South Africa	Free state	2010	NCBI
MG659755	H	Human	South Africa	Free state	2010	NCBI
MG659756	H	Human	South Africa	Free state	2010	NCBI
MG659757	H	Human	South Africa	Free state	2010	NCBI
MG659758	H	Human	South Africa	Free state	2010	NCBI
MG659759	H	Human	South Africa	Free state	2010	NCBI
MG659760	H	Human	South Africa	Free state	2010	NCBI
MG659761	H	Human	South Africa	Free state	2010	NCBI
MG659762	H	Human	South Africa	Free state	2010	NCBI
MG659763	H	Human	South Africa	Free state	2010	NCBI
MG659764	H	Human	South Africa	Free state	2010	NCBI
MG659765	H	Human	South Africa	Free state	2010	NCBI
MG659768	H	Human	South Africa	Free state	2010	NCBI
MG659769	H	Human	South Africa	Free state	2010	NCBI
MG659770	H	Human	South Africa	Free state	2010	NCBI
MG659771	H	Human	South Africa	Free state	2010	NCBI
MG659772	H	Human	South Africa	Free state	2010	NCBI
MG659773	H	Human	South Africa	Free state	2010	NCBI
MG659774	H	Human	South Africa	Free state	2010	NCBI
MG659775	H	Human	South Africa	Free state	2010	NCBI
MG659776	H	Human	South Africa	Free state	2010	NCBI
MG659777	H	Human	South Africa	Free state	2010	NCBI
MG659778	H	Human	South Africa	Free state	2010	NCBI
MG659779	H	Human	South Africa	Free state	2010	NCBI
MG659781	H	Human	South Africa	Free state	2010	NCBI
MG659782	H	Human	South Africa	Free state	2010	NCBI
MG659783	H	Human	South Africa	Free state	2010	NCBI
MG659784	H	Human	South Africa	Free state	2010	NCBI
MG659785	H	Human	South Africa	Free state	2010	NCBI
MG659786	H	Human	South Africa	Free state	2010	NCBI
MG659787	H	Human	South Africa	Free state	2010	NCBI
MG659788	H	Human	South Africa	Free state	2010	NCBI
MG659789	H	Human	South Africa	Free state	2010	NCBI
MG659790	H	Human	South Africa	Free state	2010	NCBI
MG659791	H	Human	South Africa	Free state	2010	NCBI
MG659794	H	Human	South Africa	Free state	2010	NCBI
MG659795	H	Human	South Africa	Free state	2010	NCBI
MG659796	H	Human	South Africa	Free state	2010	NCBI
MG659798	H	Human	South Africa	Free state	2010	NCBI
MG659799	H	Human	South Africa	Free state	2010	NCBI
MG659800	H	Human	South Africa	Free state	2010	NCBI

MG659801	H	Human	South Africa	Free state	2010	NCBI
MG659803	H	Human	South Africa	Free state	2010	NCBI
MG659804	H	Human	South Africa	Eastern cape	2011	NCBI
MG659805	H	Human	South Africa	Eastern cape	2011	NCBI
MG659806	H	Human	South Africa	Eastern cape	2011	NCBI
MG659807	H	Human	South Africa	Eastern cape	2011	NCBI
MG659808	H	Human	South Africa	Eastern cape	2011	NCBI
MG659810	H	Human	South Africa	Eastern cape	2011	NCBI
MG659811	H	Human	Namibia	Caprivi	2011	NCBI
MG659812	K	Cow	Kenya	Makueni	1962	NCBI
MG659813	B	Sheep	Kenya	Mombasa	1972	NCBI
MG659814	C	Human	Zimbabwe	Gatooma	1976	NCBI
MG659815	C	Sheep	Zimbabwe	Sinoia	1978	NCBI
MG659816	C	Cow	Zimbabwe	Salisbury	1978	NCBI
MG659818	N	Human	Senegal	Kedougou	1975	NCBI
MT561460	H	Sheep	Namibia	Karasburg	2010	NCBI
NC_014396	A	Human	Egypt	Sharqiya	1977	NCBI
ON060820	C	Human	Uganda	Kampala	2018	NCBI
OQ440143	L	Mosquito	Uganda	Lunyo	1955	NCBI
OR780618	C	Cow	Burundi	Ruyaga	2022	Study
OR780619	C	Cow	Burundi	Ngozi	2022	Study
OR780620	C	Cow	Burundi	Kobero	2022	Study
OR780621	C	Cow	Burundi	Kobero	2022	Study
OR780622	C	Cow	Burundi	Kobero	2022	Study
OR780623	C	Cow	Burundi	Kobero	2022	Study
OR780624	C	Cow	Burundi	Ngozi	2022	Study
PP746342	C	Cow	Rwanda	Nyarugenge	2022	Study
PP746343	C	Cow	Rwanda	Gicumbi	2022	Study
PP746344	C	Cow	Rwanda	Gicumbi	2022	Study
PP746345	C	Cow	Rwanda	Gicumbi	2022	Study
PP746346	C	Cow	Rwanda	Gicumbi	2022	Study
PP746347	C	Cow	Rwanda	Kamonyi	2022	Study
PP746348	C	Cow	Rwanda	Nyaruguru	2022	Study
PP746349	C	Cow	Rwanda	Nyanza	2022	Study
PP746350	C	Cow	Rwanda	Kamonyi	2022	Study
PP746351	C	Cow	Rwanda	Nyarugenge	2022	Study
PP746352	C	Cow	Rwanda	Muhanga	2022	Study
PP746353	C	Cow	Rwanda	Muhanga	2022	Study
PP746354	C	Cow	Rwanda	Rusizi	2022	Study
PP746355	C	Cow	Rwanda	Nyamasheke	2022	Study
PP746358	C	Cow	Rwanda	Nyarugenge	2022	Study
PP746360	C	Cow	Rwanda	Muhanga	2022	Study
PP746361	C	Cow	Rwanda	Muhanga	2022	Study
PP746362	C	Cow	Rwanda	Kamonyi	2022	Study
PP746364	C	Cow	Rwanda	Gicumbi	2022	Study
PP746365	C	Cow	Rwanda	Kamonyi	2022	Study
PP746366	C	Cow	Rwanda	Nyarugenge	2022	Study
PP746368	C	Cow	Rwanda	Gicumbi	2022	Study
PP746369	C	Cow	Rwanda	Kamonyi	2022	Study
PP746371	C	Sheep	Rwanda	Muhanga	2022	Study
PP746372	C	Cow	Rwanda	Muhanga	2022	Study
PP746373	C	Cow	Rwanda	Nyarugenge	2022	Study
PP746374	C	Cow	Rwanda	Nyarugenge	2022	Study
PP746375	C	Cow	Rwanda	Nyarugenge	2022	Study
PP746376	C	Cow	Rwanda	Nyarugenge	2022	Study
PP746378	C	Cow	Rwanda	Nyarugenge	2022	Study
PP746418	C	Human	Kenya	Wajir	2018	Study
PP746420	C	Human	Kenya	Kilifi	2007	Study
PP746428	C	Cow	Kenya	Marsabit	2018	Study
PP746429	C	Human	Kenya	Baringo	2007	Study
PP746430	C	Human	Kenya	Baringo	2007	Study

PP746431	C	Human	Kenya	Kilifi	2007	Study
PP746439	C	Human	Kenya	Baringo	2007	Study
PP746441	C	Human	Kenya	Baringo	2007	Study
PP746453	C	Human	Kenya	Isiolo	2020	Study
PP746456	C	Human	Kenya	Nairobi	2007	Study
PP746458	C	Human	Kenya	Nyandarua	2019	Study
PP746460	C	Human	Kenya	Isiolo	2020	Study
PP747234	C	Cow	Kenya	Wajir	2018	Study
PP747235	C	Cow	Kenya	Kiambu	2021	Study
PP747236	C	Cow	Rwanda	Ngoma	2018	Study
PP747238	C	Cow	Rwanda	Rulindo	2018	Study

Appendix Table S6: RVFV small segment genome accessions and metadata used in the phylogenetic and phylogeographic analysis

Table showing the sequence genomes for the large segment of RVFV providing information on the lineage, host, country, location and year of collection. The source column indicates publicly accessed sequences (NCBI) and those sequenced as part of the RVF studies in East Africa (Kenya, Rwanda and Burundi) from outbreak samples and samples archived from sporadic RVF outbreaks.

Accession	Lineage	Host	Country	Location	Year	Source
DQ380143	A	Cow	Zimbabwe	Beatrice	1974	NCBI
DQ380144	A	Cow	Madagascar	Moramanga	1979	NCBI
DQ380145	A	Sheep	Egypt	Gharbiya	1979	NCBI
DQ380146	A	Mosquito	Egypt	Sharqiya	1978	NCBI
DQ380147	A	Mosquito	Egypt	Sharqiya	1977	NCBI
DQ380150	A	Mosquito	Egypt	Sharqiya	1977	NCBI
DQ380152	A	Cow	Egypt	Asyut	1978	NCBI
DQ380153	A	Human	Egypt	Gharbiya	1978	NCBI
DQ380155	K	Sheep	Kenya	Kabete	1951	NCBI
DQ380156	K	Human	Uganda	Entebbe	1944	NCBI
DQ380158	O	Sheep	South Africa	Boshof	1951	NCBI
DQ380159	K	Cow	Zimbabwe	Salisbury	1974	NCBI
DQ380160	G	Human	Central African Republic	Bangui	1985	NCBI
DQ380161	G	Human	Central African Republic	Mbaiki	1985	NCBI
DQ380162	E	Human	Central African Republic	Ombella Mpoko	1973	NCBI
DQ380164	E	Cow	Zimbabwe	Salisbury	1978	NCBI
DQ380165	G	Bat	Guinea	Macenta	1981	NCBI
DQ380167	G	Human	Central African Republic	Bangui	1969	NCBI
DQ380168	G	Cow	Zimbabwe	Sinoia	1978	NCBI
DQ380169	C	Human	Kenya	Garissa	1998	NCBI
DQ380170	C	Human	Saudi Arabia	Jizan	2000	NCBI
DQ380171	C	Mosquito	Kenya	Ruiru	1983	NCBI
DQ380172	D	Human	Central African Republic	Ombella Mpoko	1973	NCBI
DQ380173	J	Cow	Zimbabwe	Sinoia	1974	NCBI
DQ380174	L	Cow	Zimbabwe	Salisbury	1970	NCBI
DQ380175	L	Human	South Africa	Randfontein	1975	NCBI
DQ380176	L	Cow	Kenya	Kiambu	1965	NCBI
DQ380177	N	Human	Mauritania	Kiffa	1987	NCBI
DQ380179	N	Human	Mauritania	Ayoum el Atrous	1987	NCBI
DQ380180	N	Human	Mauritania	Keur Macene	1987	NCBI
DQ380181	N	Mosquito	Burkina Faso	Fada-Ngourma	1983	NCBI
EU312106	C	Cow	Madagascar	Ivato	1991	NCBI
EU312107	C	Cow	Madagascar	Tanjombato	1991	NCBI
EU312108	C	Cow	Madagascar	Tanjombato	1991	NCBI
EU312121	M	Mosquito	Uganda	Lunyo	1955	NCBI
EU312138	A	Human	Egypt	Sharqiya	1977	NCBI

EU574057	C	Cow	Kenya	Kiambu	2007	NCBI
EU574058	C	Cow	Kenya	Nairobi	2007	NCBI
EU574059	C	Sheep	Kenya	Baringo	2007	NCBI
EU574060	C	Cow	Kenya	Embu	2007	NCBI
EU574061	C	Buffalo	Kenya	Garissa	2007	NCBI
EU574066	C	Cow	Kenya	Nairobi	2007	NCBI
EU574067	C	Sheep	Kenya	Nairobi	2007	NCBI
EU574070	C	Cow	Kenya	Nairobi	2007	NCBI
EU574071	C	Cow	Kenya	Mombassa	2007	NCBI
EU574072	C	Cow	Kenya	Muranga	2007	NCBI
EU574074	C	Cow	Kenya	Meru South	2007	NCBI
EU574076	C	Cow	Kenya	Nyeri	2007	NCBI
EU574081	C	Cow	Kenya	Maragua	2007	NCBI
EU574085	C	Cow	Kenya	Maragua	2007	NCBI
EU574087	C	Cow	Kenya	Maragua	2007	NCBI
EU709747	C	Human	South Africa	Phalaborwa	2008	NCBI
HE687302	C	Human	Mayotte	Bandraboua	2008	NCBI
HE687307	C	Human	Mayotte	Sada	2008	NCBI
HM586975	C	Human	Kenya	Garissa	2006	NCBI
HM586976	C	Human	Kenya	Garissa	2006	NCBI
HM586977	C	Human	Kenya	Kilifi	2007	NCBI
HM586978	C	Human	Kenya	Malindi	2007	NCBI
HM586979	C	Human	Kenya	Baringo	2007	NCBI
HM586980	C	Human	Kenya	Kiserian	2007	NCBI
HM586981	C	Human	Tanzania	Tanga	2007	NCBI
HM586982	C	Human	Tanzania	Dodoma	2007	NCBI
HM586983	C	Mosquito	Kenya	Garissa	2006	NCBI
HM586984	C	Mosquito	Kenya	Kilifi	2007	NCBI
JF311386	C	Human	Madagascar	Taolagnaro	2008	NCBI
JF311388	C	Human	Madagascar	Ankazobe	2008	NCBI
JF311390	C	Human	Madagascar	Antananarivo	1991	NCBI
JF311391	C	Cow	Madagascar	Antananarivo	1991	NCBI
JF311392	C	Cow	Madagascar	Miarinarivo	2008	NCBI
JF311393	C	Cow	Madagascar	Antsirabe	2008	NCBI
JF326198	C	Mosquito	Kenya	Garissa	2007	NCBI
JF326199	C	Mosquito	Kenya	Garissa	2006	NCBI
JF326200	C	Mosquito	Kenya	Garissa	2006	NCBI
JF326201	C	Mosquito	Kenya	Garissa	2006	NCBI
JF326203	C	Human	Tanzania	Dodoma	2007	NCBI
JF784388	L	Sheep	South Africa	Free State	1974	NCBI
JQ820472	C	Human	Sudan	White Nile	2007	NCBI
JQ820473	C	Human	Sudan	Gezira	2010	NCBI
JQ820474	C	Human	Sudan	Gezira	2010	NCBI
JQ820475	C	Human	Sudan	Gezira	2010	NCBI
JQ820476	C	Human	Sudan	Gezira	2010	NCBI
JQ820477	C	Human	Sudan	Gezira	2010	NCBI
JQ820478	C	Human	Sudan	Khartoum	2007	NCBI
JQ820481	C	Human	Sudan	Gezira	2010	NCBI
JQ820482	C	Human	Sudan	Khartoum	2007	NCBI
JQ840746	C	Human	Sudan	White Nile	2007	NCBI
KM210508	K	Camel	Mauritania	Aoujeft	2010	NCBI
KU978774	C	Human	Kenya	Garissa	1998	NCBI
KX096940	C	Mosquito	Kenya	Garissa	2006	NCBI
KX096943	C	Mosquito	Saudi Arabia	Aseer	2001	NCBI
KX944803	H	Cow	South Africa	Sterkfontein	2010	NCBI
KX944804	H	Cow	South Africa	Groenhof	2009	NCBI
KX944805	H	Sheep	South Africa	Bultfontein	2010	NCBI
KX944806	H	Sheep	South Africa	Bultfontein	2010	NCBI
KX944807	O	Mosquito	South Africa	Free State	1955	NCBI
KX944808	L	Cow	South Africa	Natal	1975	NCBI
KX944809	H	Sheep	South Africa	Heldefontein	2010	NCBI
KX944811	H	Sheep	South Africa	Uppington	2009	NCBI

KX944813	H	Sheep	South Africa	Bultfontein	2010	NCBI
KX944814	H	Cow	South Africa	Upington	2009	NCBI
KX944815	H	Cow	South Africa	Pretoria	2010	NCBI
KX944816	H	Sheep	South Africa	Middelburg	2010	NCBI
KX944817	C	Buffalo	South Africa	Hoedspruit	2008	NCBI
KX944818	C	Buffalo	South Africa	Mbombela	2008	NCBI
KX944821	L	Human	South Africa	Eshowe	1974	NCBI
KX944822	C	Cow	South Africa	Modderfontein	2009	NCBI
KX944823	C	Cow	South Africa	Centurion	2008	NCBI
KX944824	C	Cow	South Africa	Centurion	2008	NCBI
KX944825	C	Cow	South Africa	Centurion	2008	NCBI
KY196487	H	Human	South Africa	Northern cape	2010	NCBI
KY196488	H	Human	South Africa	Free state	2010	NCBI
KY196489	H	Human	South Africa	Northern cape	2010	NCBI
KY196491	H	Human	South Africa	Free state	2010	NCBI
KY196492	H	Human	South Africa	Free state	2010	NCBI
KY196493	H	Human	South Africa	Free state	2010	NCBI
KY196494	H	Human	South Africa	Free state	2010	NCBI
KY196495	H	Human	South Africa	Free state	2010	NCBI
KY196496	H	Human	South Africa	Free state	2010	NCBI
KY196497	H	Human	South Africa	Free state	2010	NCBI
KY196498	H	Human	South Africa	Eastern cape	2011	NCBI
KY196499	H	Human	South Africa	Eastern cape	2011	NCBI
KY196500	H	Human	Namibia	Caprivi	2004	NCBI
KY196501	C	Buffalo	South Africa	Mpumalanga	2008	NCBI
KY196502	C	Buffalo	South Africa	Mpumalanga	2008	NCBI
KY196503	C	Cow	South Africa	Bela-Bela	2008	NCBI
KY196505	C	Cow	South Africa	Northern cape	2009	NCBI
KY196506	H	Human	South Africa	Eastern cape	2009	NCBI
KY196507	I	Mosquito	South Africa	Free State	1955	NCBI
KY196508	I	Mosquito	South Africa	Free State	1956	NCBI
KY196509	L	Mosquito	South Africa	Standerton	1971	NCBI
KY196510	F	Mosquito	South Africa	Natal	1981	NCBI
KY196511	C	Buffalo	South Africa	Limpopo	1999	NCBI
KY366328	A	Human	Mauritania	Maghta Lahjar	2015	NCBI
MG659905	M	Mosquito	South Africa	Free State	1955	NCBI
MG659906	H	Human	South Africa	Northern cape	2010	NCBI
MG659907	H	Human	South Africa	Free state	2010	NCBI
MG659910	H	Human	South Africa	Free state	2010	NCBI
MG659914	H	Human	South Africa	Free state	2010	NCBI
MG659917	H	Human	South Africa	Eastern cape	2010	NCBI
MG659920	H	Human	South Africa	Free state	2010	NCBI
MG659924	H	Human	South Africa	Free state	2010	NCBI
MG659926	H	Human	South Africa	Free state	2010	NCBI
MG659928	H	Human	South Africa	Free state	2010	NCBI
MG659930	H	Human	South Africa	Free state	2010	NCBI
MG659931	H	Human	South Africa	Free state	2010	NCBI
MG659933	H	Human	South Africa	Free state	2010	NCBI
MG659934	H	Human	South Africa	Free state	2010	NCBI
MG659936	H	Human	South Africa	Free state	2010	NCBI
MG659946	H	Human	South Africa	Free state	2010	NCBI
MG659947	H	Human	South Africa	Free state	2010	NCBI
MG659951	H	Human	South Africa	Free state	2010	NCBI
MG659953	H	Human	South Africa	Free state	2010	NCBI
MG659954	H	Human	South Africa	Free state	2010	NCBI
MG659955	H	Human	South Africa	Free state	2010	NCBI
MG659956	H	Human	South Africa	Free state	2010	NCBI
MG659958	H	Human	South Africa	Free state	2010	NCBI
MG659962	H	Human	South Africa	Free state	2010	NCBI
MG659963	H	Human	South Africa	Free state	2010	NCBI
MG659964	H	Human	South Africa	Free state	2010	NCBI
MG659965	H	Human	South Africa	Free state	2010	NCBI

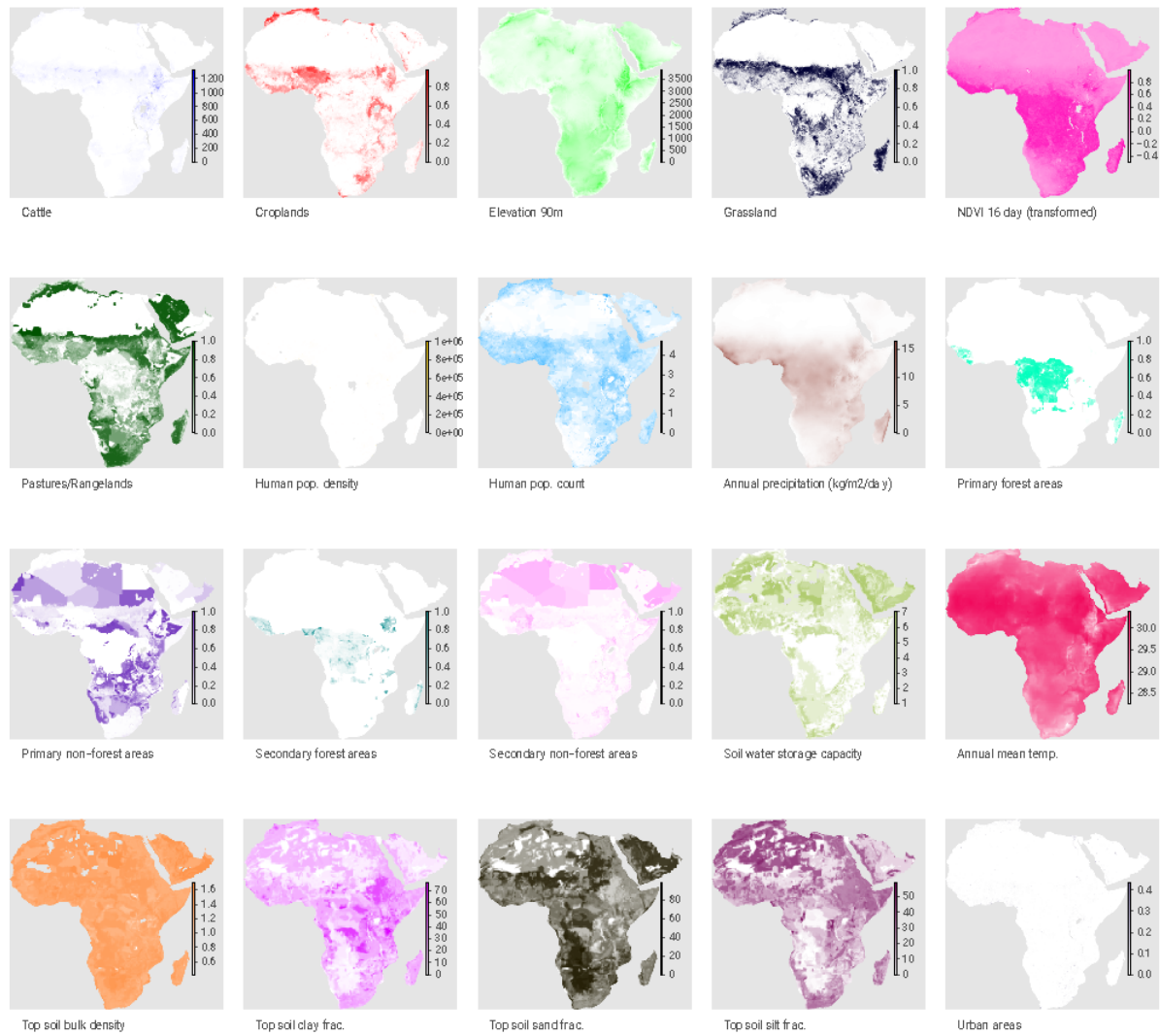
MG659966	H	Human	South Africa	Free state	2010	NCBI
MG659968	H	Human	South Africa	Free state	2010	NCBI
MG659970	H	Human	South Africa	Free state	2010	NCBI
MG659971	H	Human	South Africa	Free state	2010	NCBI
MG659973	H	Human	South Africa	Free state	2010	NCBI
MG659977	H	Human	South Africa	Eastern cape	2011	NCBI
MG659981	H	Human	South Africa	Eastern cape	2011	NCBI
MG659982	H	Human	South Africa	Eastern cape	2011	NCBI
MG659984	K	Cow	Kenya	Makueni	1962	NCBI
MG659985	B	Sheep	Kenya	Mombasa	1972	NCBI
MG659986	C	Human	Zimbabwe	Gatooma	1976	NCBI
MG659987	C	Sheep	Zimbabwe	Sinoia	1978	NCBI
MG659988	C	Cow	Zimbabwe	Salisbury	1978	NCBI
MG659989	M	Mosquito	Uganda	Lunyo	1955	NCBI
MG659990	N	Human	Senegal	Kedougou	1975	NCBI
MG953424	C	Human	Uganda	Kabale	2016	NCBI
MG953425	C	Human	Uganda	Kamuganguzi	2016	NCBI
MG953426	C	Human	Uganda	Kabale	2016	NCBI
MG972974	C	Human	Uganda	Mityana	2017	NCBI
MG972977	C	Human	Uganda	Kiboga	2017	NCBI
MG972980	C	Human	Uganda	Sembabule	2017	NCBI
MT561459	H	Sheep	Namibia	Karasburg	2010	NCBI
MT561462	C	Springbok	Namibia	Oshikoto	2011	NCBI
NC_01439	A	Human	Egypt	Sharqiya	1977	NCBI
5						
OM744370	C	Cow	Kenya	Kiambu	2021	NCBI
OM744371	C	Cow	Kenya	Kiambu	2021	NCBI
ON060771	C	Human	Uganda	Kampala	2019	NCBI
ON060774	C	Human	Uganda	Kampala	2019	NCBI
ON060786	K	Human	Uganda	Kampala	2019	NCBI
ON060792	C	Human	Uganda	Kampala	2018	NCBI
ON060798	C	Human	Uganda	Kampala	2018	NCBI
ON060801	C	Human	Uganda	Kampala	2018	NCBI
ON060807	C	Human	Uganda	Kampala	2018	NCBI
ON060810	C	Human	Uganda	Kampala	2018	NCBI
ON060812	C	Human	Uganda	Kampala	2018	NCBI
ON060821	C	Human	Uganda	Kampala	2018	NCBI
ON060824	C	Human	Uganda	Kampala	2020	NCBI
ON060827	C	Human	Uganda	Kampala	2020	NCBI
ON060830	C	Human	Uganda	Kampala	2020	NCBI
ON060833	C	Human	Uganda	Kampala	2020	NCBI
ON060835	C	Human	Uganda	Kampala	2020	NCBI
ON060838	C	Human	Uganda	Kampala	2020	NCBI
OR780613	C	Cow	Burundi	Ruyaga	2022	Study
OR780614	C	Cow	Burundi	Ngozi	2022	Study
OR780615	C	Cow	Burundi	Kobero	2022	Study
OR780617	C	Cow	Burundi	Kobero	2022	Study
PP746379	C	Cow	Rwanda	Nyarugenge	2022	Study
PP746380	C	Cow	Rwanda	Gicumbi	2022	Study
PP746381	C	Cow	Rwanda	Gicumbi	2022	Study
PP746382	C	Cow	Rwanda	Gicumbi	2022	Study
PP746383	C	Cow	Rwanda	Gicumbi	2022	Study
PP746384	C	Cow	Rwanda	Kamonyi	2022	Study
PP746386	C	Cow	Rwanda	Nyanza	2022	Study
PP746387	C	Cow	Rwanda	Kamonyi	2022	Study
PP746389	C	Cow	Rwanda	Muhanga	2022	Study
PP746392	C	Cow	Rwanda	Nyarugenge	2022	Study
PP746393	C	Cow	Rwanda	Nyamasheke	2022	Study
PP746395	C	Cow	Rwanda	Nyarugenge	2022	Study
PP746398	C	Cow	Rwanda	Muhanga	2022	Study
PP746399	C	Cow	Rwanda	Muhanga	2022	Study
PP746400	C	Cow	Rwanda	Kamonyi	2022	Study

PP746401	C	Cow	Rwanda	Nyarugenge	2022	Study
PP746402	C	Cow	Rwanda	Gicumbi	2022	Study
PP746404	C	Cow	Rwanda	Nyarugenge	2022	Study
PP746407	C	Cow	Rwanda	Nyarugenge	2022	Study
PP746408	C	Sheep	Rwanda	Muhanga	2022	Study
PP746409	C	Cow	Rwanda	Muhanga	2022	Study
PP746410	C	Cow	Rwanda	Nyarugenge	2022	Study
PP746411	C	Cow	Rwanda	Nyarugenge	2022	Study
PP746412	C	Cow	Rwanda	Nyarugenge	2022	Study
PP746414	C	Cow	Rwanda	Nyarugenge	2022	Study
PP746421	C	Human	Kenya	Wajir	2018	Study
PP746432	C	Cow	Kenya	Marsabit	2018	Study
PP746433	C	Human	Kenya	Baringo	2007	Study
PP746434	C	Human	Kenya	Baringo	2007	Study
PP746435	C	Human	Kenya	Kilifi	2007	Study
PP746442	C	Human	Kenya	Baringo	2007	Study
PP746444	C	Human	Kenya	Baringo	2007	Study
PP746461	C	Human	Kenya	Isiolo	2020	Study
PP746464	C	Human	Kenya	Nairobi	2007	Study
PP746465	C	Human	Kenya	Baringo	2007	Study
PP746466	C	Human	Kenya	Nyandarua	2019	Study
PP746468	C	Human	Kenya	Isiolo	2020	Study
PP747239	C	Cow	Kenya	Wajir	2018	Study
PP747240	C	Cow	Kenya	Kiambu	2021	Study
PP747241	C	Cow	Rwanda	Ngoma	2018	Study
PP747243	C	Cow	Rwanda	Rulindo	2018	Study

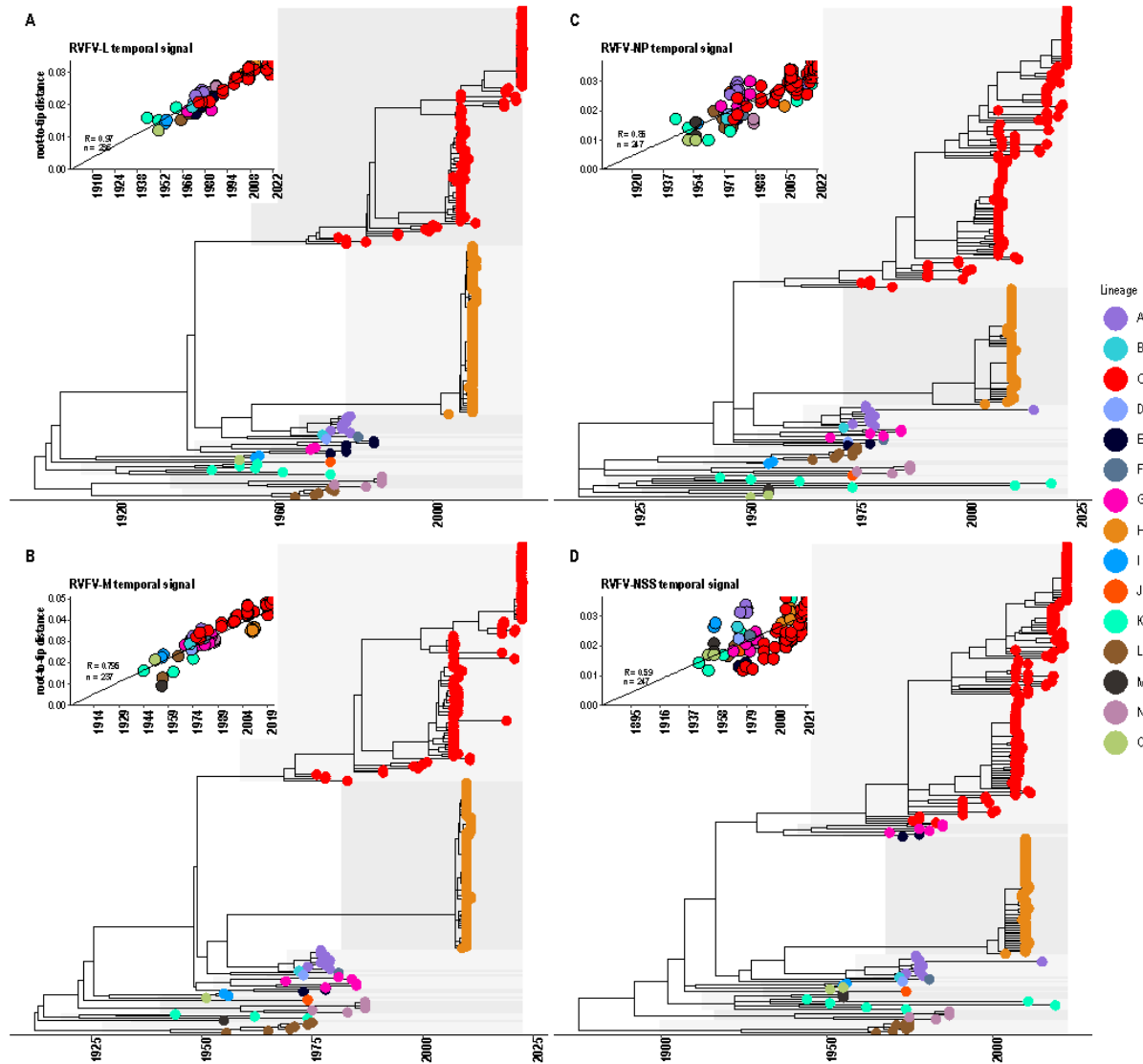
Appendix Table S7. Environmental raster data sources used as background for landscape phylogeography associations.

Environmental raster	Source	URL
<i>Annual mean precipitation</i>	WorldClim database, version 2.1 (bioclimatic variable: “bio1”)	https://www.worldclim.org
<i>Annual mean precipitation</i>	WorldClim database, version 2.1 (bioclimatic variable: “bio12”)	https://www.worldclim.org
<i>Elevation</i>	SRTM (Shuttle Radar Topography Mission)	https://srtm.csi.cgiar.org/
<i>Land cover raster (Croplands, Pastures, Primary Forest areas, Primary non-forest areas, Secondary Forest areas, Secondary non-forest areas, Urban areas)</i>	IGBP (International Geosphere Biosphere Programme) – categorical raster	http://www.igbp.net
<i>Normalized Difference Vegetation Index (NDVI)</i>	Normalized Difference Vegetation Index, NASA Earth Observation database	https://modis.gsfc.nasa.gov/data/
<i>Cattle (Gridded Livestock density (Number of cattle per square kilometre))</i>	FAO (Food and Agriculture Organization of the United Nations)	https://www.fao.org/livestock-systems/global-distributions/cattle/en/

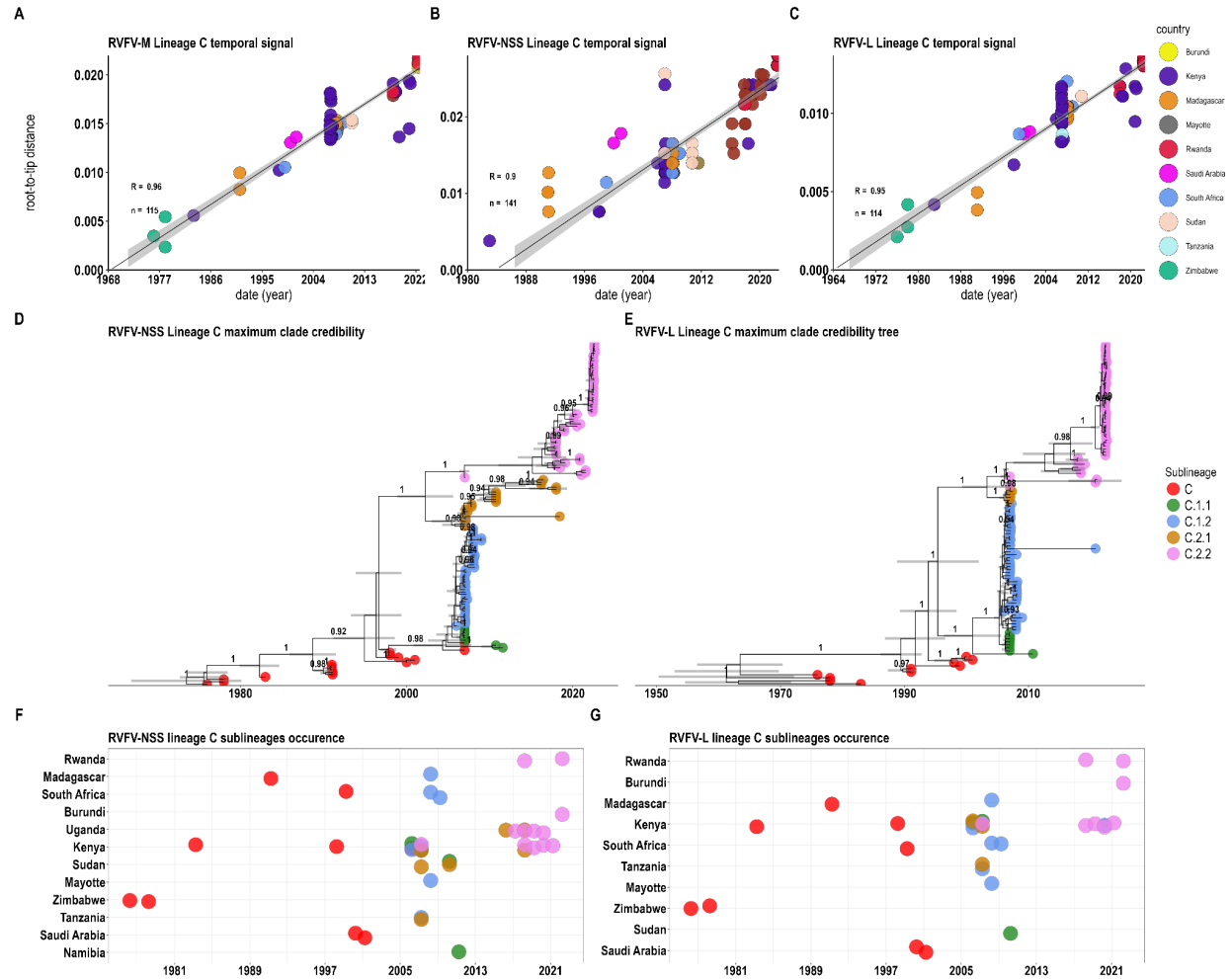
<i>Grassland</i>	ESA (European Space Agency)	https://esa-worldcover.org/en/data-access
<i>Soils (Topsoil Bulk density, available water storage capacity, topsoil clay fraction, topsoil sand fraction, topsoil silt fraction)</i>	Regridded Harmonized World Soil Database v1.2	https://daac.ornl.gov/SOILS/guides/HWSD.html
<i>Population</i>	NASA Earthdata	https://www.earthdata.nasa.gov/data/catalog/sedac-ciesin-sedac-gpww4-apct-wpp-2015-r11-4.11



Appendix Figure S2. Environmental variables tested for their impact on the dispersal of Rift Valley fever virus in Africa.

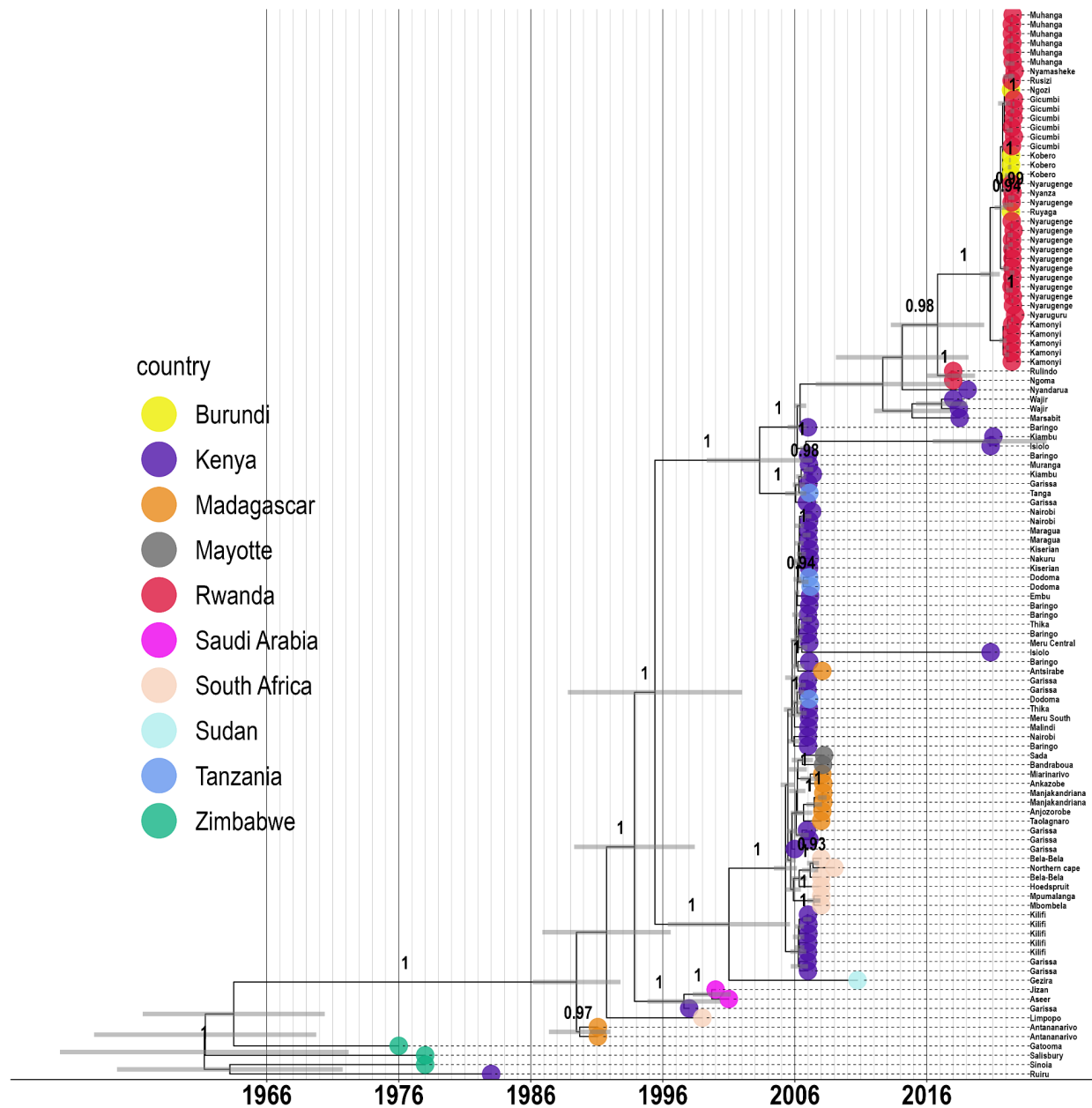


Appendix Figure S3. Time-resolved maximum likelihood (ML) phylogenetic trees of RVFV genetic sequence data. (A) ML phylogenetic tree of the large segment. (B) ML phylogenetic tree of the medium segment. (C) ML phylogenetic tree of the nucleocapsid gene sequence data and (D) ML phylogenetic tree of the non-structural gene sequence data. Root-to-tip regression plots showing the temporal signal of the sequence data as a regression of genetic distance and time in years. The temporal signal was assessed using linear regression function in R and visualised with the R package “ggplot2”. Time-resolved ML phylogenetic trees for the sequence data (RVFV-L, $n=236$, RVFV-M, $n=237$, RVFV-NP, $n=247$ and RVFV-NSs, $n=247$). Trees were inferred from the sequence data using GTR+I+G4 as the substitution model with 1,000 replicates as bootstrapping procedure. The phylogenetic tree was visualised using the R package “ggtree”²⁵.



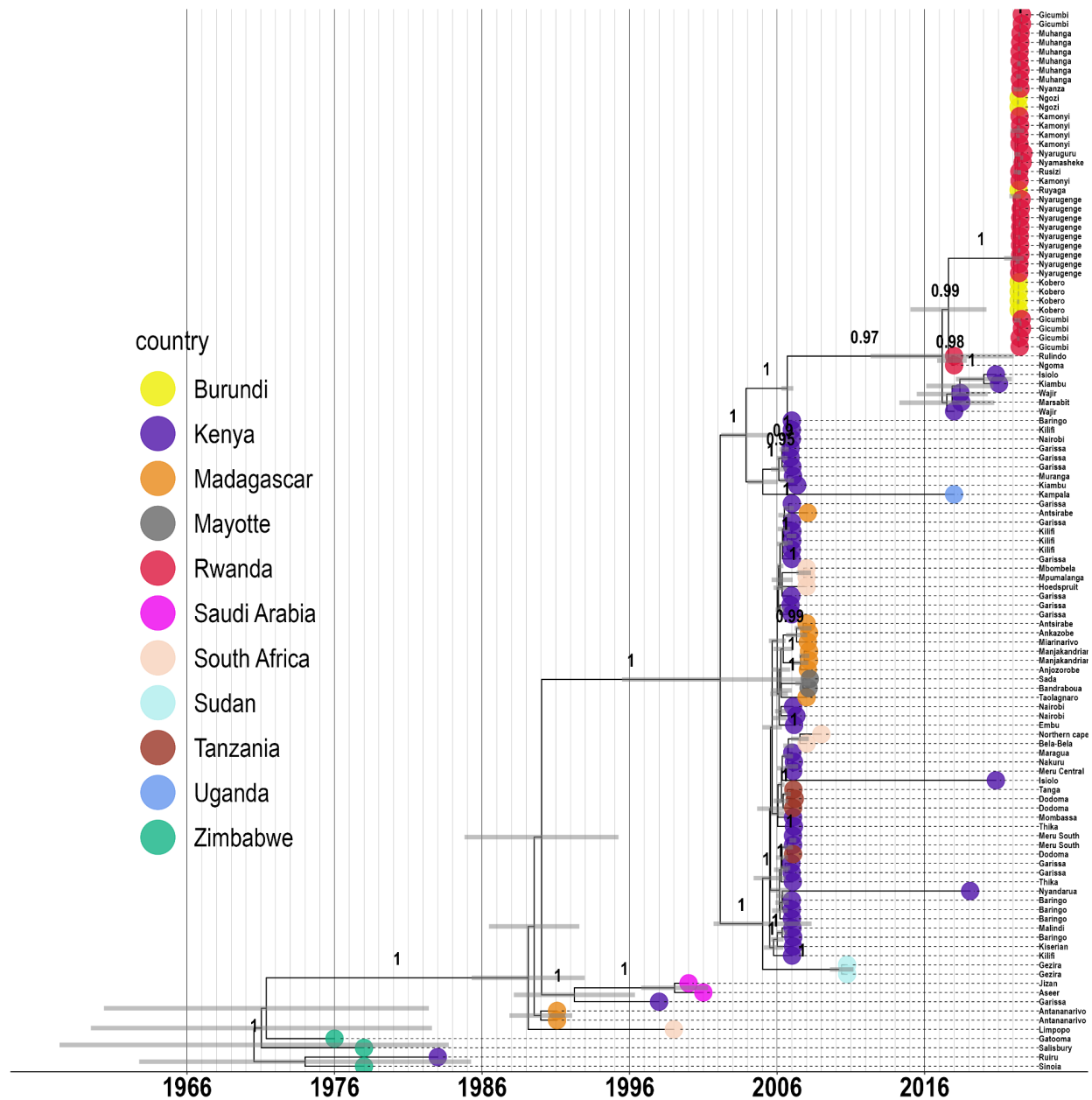
Appendix Figure S4. Time-resolved maximum likelihood (ML) phylogenetic trees and occurrence of RVFV lineage C sub-lineages. The temporal signals of the genetic sequences assessed by regression of the root-to-tip genetic distance and time (years) for (A) medium segment, (B) non-structural gene sequences and (C) large segment. Time-resolved ML phylogenetic trees for the (D) non-structural gene sequences, $n=141$ and (E) the large segment ($n=114$), here highlighting the clustering of the different sub-lineages of the lineage C virus strains for (D) non-structural gene sequences and (E) the large segment. Occurrence of different sub-lineages of the lineage C in Africa and Saudi Arabia for (F) non-structural gene and (G) the large segment sequences since the collection of the first sample in 1976 (Zimbabwe) to 2023 (Rwanda).

RVFV-L



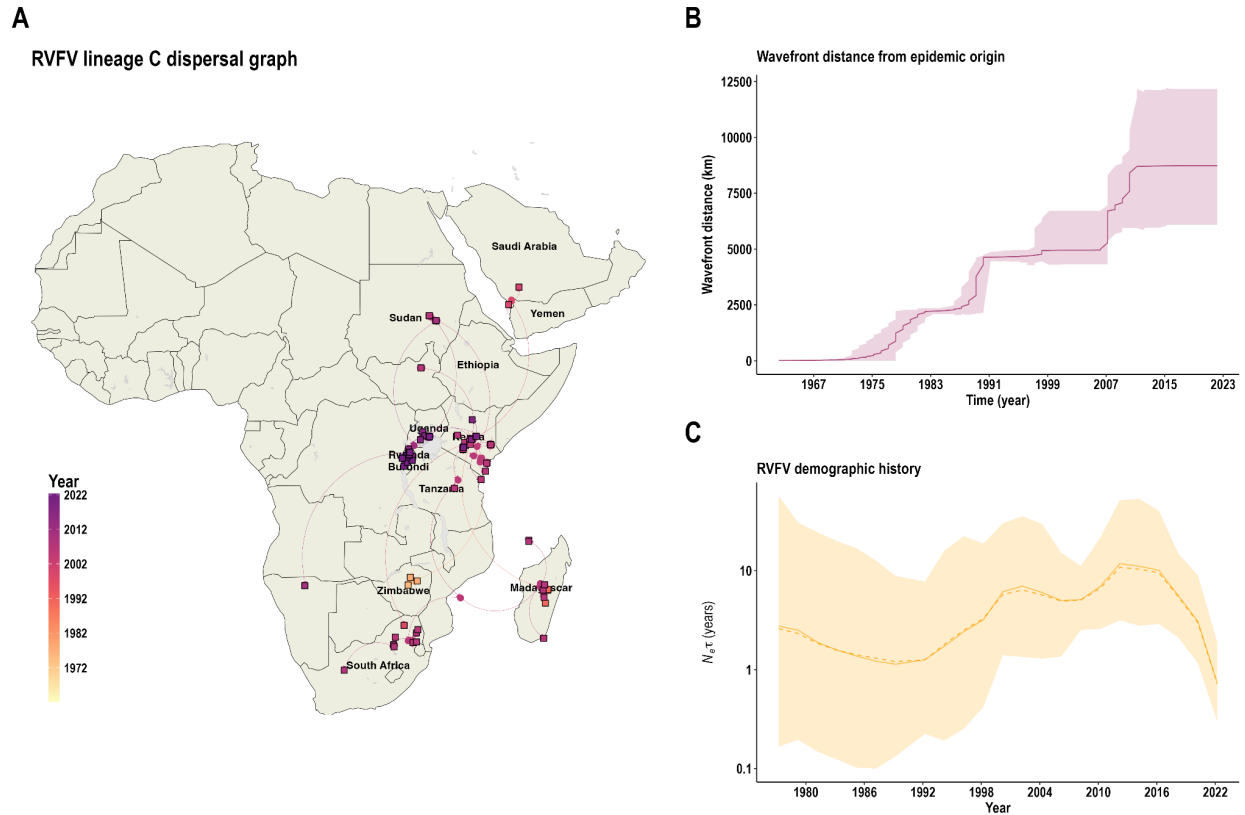
Appendix Figure S5. Evolutionary analysis of lineage C using the large segment sequences. Bayesian phylogenetic analysis of RVFV lineage C large segment genomic sequences (n=114) showing the maximum clade credibility (MCC) tree for lineage (cluster/clade) C. Posterior probabilities are shown on branches, with 95% highest posterior density (HPD) intervals represented as grey bars and clade posterior values are shown on branches. The tips of the tree are coloured by country origin of the samples and labelled by sampling location names.

RVFV-M

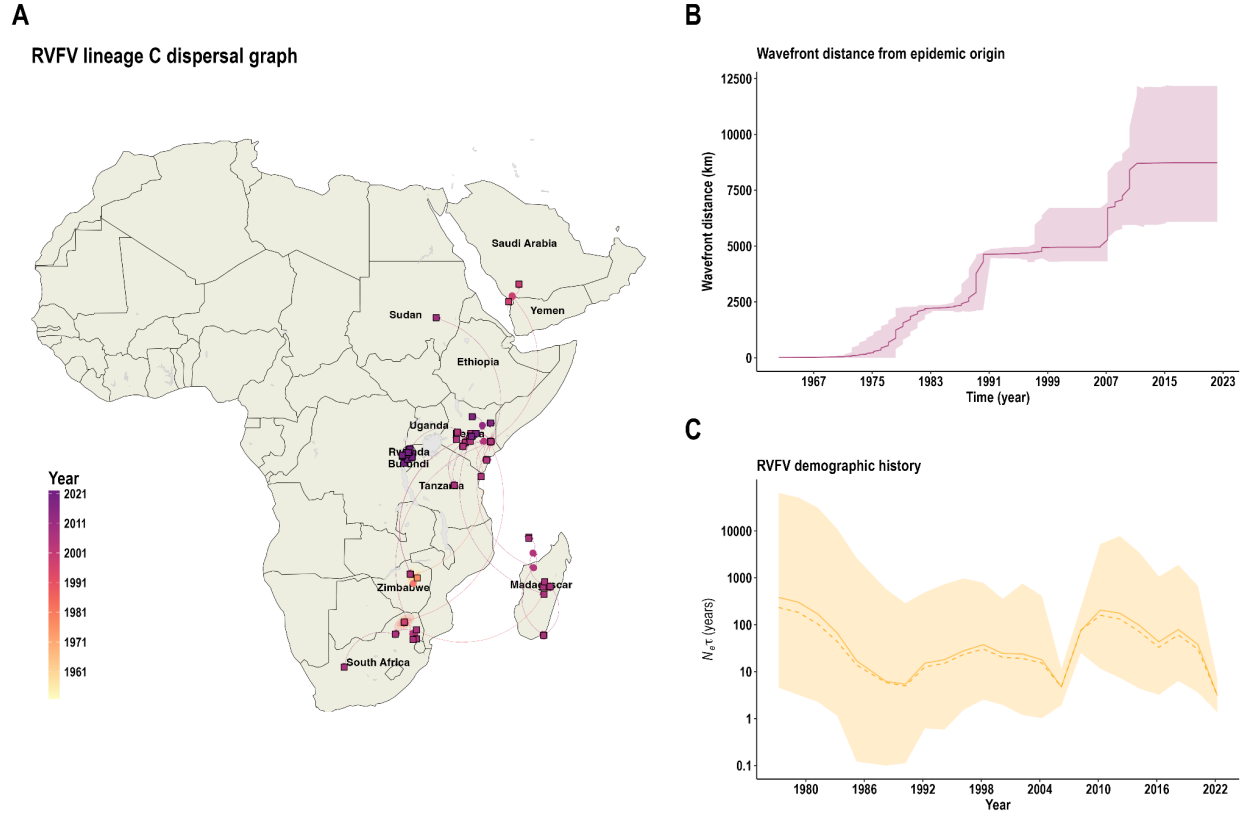


Appendix Figure S6. Evolutionary analysis of lineage C using RVFV medium segment sequences. Bayesian phylogenetic analysis of RVFV lineage C medium segment genomic sequences (n=115) showing the maximum clade credibility (MCC) tree for lineage (cluster/clade) C. Posterior probabilities are shown on branches, with 95% highest posterior density (HPD) intervals represented as grey bars clade posterior values are shown on branches. The tips of the tree are coloured by country origin of the samples and labelled by sampling location names.

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Appendix Figure S8. Spatiotemporal diffusion and dispersal dynamics of RVFV lineage C in Africa inferred using large segment sequences. Dispersal history of RVFV lineage C summarised by a maximum clade credibility (MCC) tree retrieved and annotated from 1,000 posterior trees sampled from the posterior distribution of the continuous phylogeographic analysis (**A**). Nodes of the trees are coloured according to the time of occurrence, and oldest nodes are plotted on top of youngest ones. RVFV lineage C dispersal statistics showing the evolution of the maximal wavefront distance from the epidemic origin (median 827 km; 95% HPD: 0 – 8127 km) (**B**) and weighted lineage dispersal velocity through time (**C**). We also display the effective virus population size (log transformed) estimated over the last 50 years (median 17; 95% HPD: 2 - 83) (**D**). The shaded areas correspond to the 95% credible regions, the solid lines represent the median values of estimates of interest (maximal wavefront distance, weighted lineage dispersal velocity, and effective viral population size), and the dashed line in the effective population size indicates the median estimate.



Appendix Figure S9. Spatiotemporal diffusion and dispersal dynamics of RVFV lineage C in Africa inferred using non-structural gene sequences. Dispersal history of RVFV lineage C summarised by a maximum clade credibility (MCC) tree retrieved and annotated from 1,000 posterior trees sampled from the posterior distribution of the continuous phylogeographic analysis (**A**). Nodes of the trees are coloured according to the time of occurrence, and oldest nodes are plotted on top of youngest ones. RVFV lineage C dispersal statistics showing the evolution of the maximal wavefront distance from the epidemic origin (median 438 km; 95% HPD: 0 – 8734 km) (**B**) and weighted lineage dispersal velocity through time (**C**). We also display the effective virus population size (log transformed) estimated over the last 50 years (median 4; 95% HPD: 2 - 11) (**D**). The shaded areas correspond to the 95% credible regions, the solid lines represent the median values of estimates of interest (maximal wavefront distance, weighted lineage dispersal velocity, and effective viral population size), and the dashed line in the effective population size indicates the median estimate.

Appendix Table S8. Association between environmental conditions and the dispersal location of inferred RVFV lineage C branches (viral lineages).

The results are Bayes factor (BF) supports for association between environmental raster values and tree node locations. The results are based on the analysis of 300 posterior trees obtained from the continuous phylogeographic analysis conducted for each of three RVFV genome segments. Following the scale of interpretation of Kass and Raftery²⁶, a BF value >20 is considered a strong support (in bold). “NDVI” stands for “normalised difference vegetative index”.

Environmental variable	Tendency of RVFV lineages to avoid circulating within specific environmental conditions			Tendency of RVFV lineages to preferentially circulate within specific environmental conditions		
	L segment	M segment	NS segment	L segment	M segment	NS segment
Cattle	0.5	0.3	0.1	1.9	3.5	17.7
Croplands	0.0	0.0	0.0	99.0	74.0	>99
Elevation	0.4	0.4	0.2	2.3	2.5	4.8
Grassland	3.2	2.1	1.9	0.3	0.5	0.5
NDVI	0.1	0.2	0.5	15.7	6.5	1.9
Pastures	2.7	1.7	1.2	0.4	0.6	0.8
Human population density	0.0	0.0	0.0	59.0	41.9	>99
Human population count	0.0	0.0	0.0	>99	>99	>99
Precipitation	0.4	0.5	1.3	2.3	2.0	0.7
Primary forests	0.2	0.2	1.9	5.7	4.6	0.5
Primary non-forests	2.2	3.2	5.0	0.4	0.3	0.2
Secondary forests	0.4	0.2	0.3	2.8	4.1	3.9
Secondary non-forests	1.4	1.8	0.7	0.7	0.6	1.5
Soil water storage capacity	17.7	41.9	11.5	0.1	0.0	0.1
Temperature	6.3	3.5	11.5	0.2	0.3	0.1
Topsoil bulk density	5.8	7.6	2.7	0.2	0.1	0.4
Topsoil clay	0.2	0.0	0.1	4.2	26.3	19.0
Topsoil sand	1.9	4.6	41.9	0.5	0.2	0.0
Topsoil silt	2.7	1.0	0.4	0.4	1.0	2.8
Urban areas	0.0	0.0	0.0	49.0	59.0	>99

Appendix Table S9. Impact of environmental factors on the diffusion velocity of RVFV lineages.

The output is based on 300 posterior trees obtained from continuous phylogeographic inference. The suffixes “R” and “C” indicate whether the considered environmental raster was treated as a conductance (“C”) or resistance (“R”) factor, and “*k*” is the rescaling parameter applied to transform the original raster values. The regression coefficients and *Q* statistic values are reported as median estimates and 95% HPD interval values (in square brackets). Bayes factor (BF) supports are also reported when the probability of *Q* statistic is greater than zero ($p(Q>0) > 0.9$). Applying the guidelines by Kass and Raftery²⁶, a BF value >20 is considered a strong support (in bold) for an association between the environmental distances and dispersal durations associated with phylogenetic branches.

Segment	<i>k</i>	Environmental factor	Regression coefficient	<i>Q</i> statistic	$p(Q) > 0$	BF
Large	10	Cattle (R)	0.020 [0.005, 0.046]	0.011 [0.002, 0.025]	1.00	2.3
	100	Cattle (R)	0.045 [0.012, 0.101]	0.031 [0.007, 0.087]	1.00	3.1
	1000	Cattle (R)	0.041 [0.011, 0.105]	0.03 [0.000, 0.092]	0.99	2.4
	10	Croplands (R)	0.018 [0.004, 0.042]	0.007 [0.000, 0.018]	0.99	0.7
	10	NDVI (R)	0.021 [0.003, 0.045]	0.010 [-0.009, 0.036]	0.92	1.0
	10	Pastures (C)	0.022 [0.006, 0.048]	0.012 [0.002, 0.025]	1.00	3.1
	100	Pastures (C)	0.025 [0.007, 0.051]	0.015 [0.004, 0.032]	1.00	4.0
	1000	Pastures (C)	0.026 [0.008, 0.051]	0.015 [0.004, 0.032]	1.00	4.3
	10	Population (R)	0.050 [0.016, 0.096]	0.009 [0.000, 0.022]	0.987	3.3
	100	Population (R)	0.050 [0.015, 0.096]	0.009 [0.000, 0.025]	0.981	3.4

1000	Population (R)	0.051 [0.016, 0.097]	0.009 [0.000, 0.025]	0.980	3.3
10	Precipitation (R)	0.017 [0.004, 0.039]	0.007 [-0.008, 0.022]	0.92	1.0
10	Primary forest (C)	0.044 [0.017, 0.080]	0.002 [-0.001, 0.007]	0.943	1.9
100	Primary forest (C)	0.045 [0.016, 0.079]	0.003 [-0.001, 0.010]	0.932	2.2
10	Primary non-forests (R)	0.041 [0.016, 0.077]	0.030 [0.009, 0.062]	1.00	11.5
100	Primary non-forests (R)	0.053 [0.021, 0.108]	0.042 [0.001, 0.093]	0.99	17.8
1000	Primary non-forests (R)	0.053 [0.022, 0.110]	0.043 [0.013, 0.107]	0.99	17.8
10	Secondary forests (C)	0.045 [0.015, 0.079]	0.004 [0.000, 0.008]	0.98	2.1
100	Secondary forests (C)	0.048 [0.016, 0.079]	0.007 [-0.003, 0.014]	0.94	2.5
10	Soil water storage capacity (C)	0.022 [0.008, 0.046]	0.012 [0.000, 0.025]	0.99	2.5
100	Soil water storage capacity (C)	0.026 [0.009, 0.053]	0.016 [0.000, 0.035]	0.99	2.6
1000	Soil water storage capacity (C)	0.026 [0.009, 0.054]	0.016 [0.000, 0.036]	0.98	2.6
10	Topsoil bulk density (R)	0.009 [0.001, 0.026]	0.000 [0.000, 0.001]	0.97	1.4
10	Temperature (R)	0.048 [0.017, 0.087]	0.007 [0.000, 0.016]	0.99	4.2
100	Temperature (R)	0.049 [0.018, 0.088]	0.008 [0.000, 0.018]	0.99	5.0
1000	Temperature (R)	0.049 [0.017, 0.088]	0.008 [0.000, 0.018]	0.99	5.1
100	Topsoil bulk density (R)	0.009 [0.001, 0.026]	0.000 [0.000, 0.001]	0.97	1.4
1000	Topsoil bulk density (R)	0.009 [0.001, 0.026]	0.000 [0.000, 0.001]	0.97	1.4
10	Topsoil clay (R)	0.023 [0.005, 0.051]	0.014 [0.004, 0.031]	1.00	3.8
100	Topsoil clay (R)	0.029 [0.008, 0.066]	0.019 [0.005, 0.043]	1.00	4.6
1000	Topsoil clay (R)	0.030 [0.008, 0.068]	0.020 [0.006, 0.045]	1.00	4.7
10	Topsoil sand (C)	0.017 [0.003, 0.042]	0.008 [0.001, 0.019]	1.00	2.7
100	Topsoil sand (C)	0.020 [0.004, 0.048]	0.010 [0.002, 0.025]	1.00	3.1
1000	Topsoil sand (C)	0.020 [0.004, 0.048]	0.011 [0.002, 0.026]	1.00	3.1
10	Urban areas (R)	0.010 [0.001, 0.030]	0.001 [0.000, 0.004]	1.00	1.5
100	Urban areas (R)	0.029 [0.003, 0.087]	0.017 [-0.003, 0.070]	0.97	1.3
10	Cattle (R)	0.030 [0.002, 0.081]	0.012 [0.000, 0.030]	0.98	1.6
100	Cattle (R)	0.048 [0.008, 0.103]	0.027 [-0.025, 0.081]	0.93	1.9
10	Croplands (R)	0.032 [0.003, 0.077]	0.013 [-0.008, 0.040]	0.95	0.5
10	Elevation (R)	0.032 [0.008, 0.085]	0.016 [-0.004, 0.041]	0.95	1.1
100	Elevation (R)	0.042 [0.009, 0.096]	0.024 [-0.008, 0.065]	0.94	1.0

Medium	1000	Elevation (R)	0.043 [0.011, 0.096]	0.026 [-0.010, 0.066]	0.93	1.0
	10	NDVI (R)	0.057 [0.017, 0.145]	0.036 [-0.011, 0.103]	0.96	1.2
	100	NDVI (R)	0.058 [0.017, 0.155]	0.041 [-0.013, 0.122]	0.93	1.0
	1000	NDVI (R)	0.058 [0.015, 0.154]	0.041 [-0.014, 0.122]	0.93	1.0
	10	Pastures (C)	0.039 [0.008, 0.111]	0.020 [0.001, 0.058]	1.00	3.5
	100	Pastures (C)	0.044 [0.014, 0.134]	0.025 [0.001, 0.075]	1.00	5.2
	1000	Pastures (C)	0.044 [0.011, 0.133]	0.025 [0.001, 0.077]	1.00	5.2
	10	Population (R)	0.016 [0.001, 0.065]	0.001 [-0.001, 0.004]	0.96	0.8
	10	Precipitation (R)	0.042 [0.008, 0.109]	0.023 [-0.017, 0.064]	0.95	1.2
	10	Primary forest (C)*	0.109 [0.058, 0.176]	0.013 [0.003, 0.022]	0.98	5.3
	100	Primary forest (C)*	0.115 [0.056, 0.174]	0.018 [0.004, 0.032]	0.98	7.1
	1000	Primary forest (C)*	0.120 [0.066, 0.185]	0.022 [0.005, 0.037]	0.98	8.3
	10	Primary non-forests (R)	0.076 [0.013, 0.172]	0.055 [0.004, 0.130]	1.00	7.3
	100	Primary non-forests (R)	0.119 [0.031, 0.254]	0.098 [0.022, 0.235]	1.00	8.1
	1000	Primary non-forests (R)	0.125 [0.031, 0.265]	0.102 [0.021, 0.246]	1.00	8.1
	10	Secondary forests (C)	0.113 [0.056, 0.182]	0.016 [0.004, 0.028]	1.0	5.1
	100	Secondary forests (C)	0.122 [0.059, 0.186]	0.024 [-0.004, 0.049]	0.96	6.1
	10	Soil water storage capacity (C)	0.045 [0.013, 0.113]	0.026 [-0.002, 0.070]	0.99	2.0
	100	Soil water storage capacity (C)	0.054 [0.013, 0.127]	0.033 [0.001, 0.098]	0.99	2.1
	1000	Soil water storage capacity (C)	0.054 [0.014, 0.130]	0.034 [0.000, 0.101]	0.99	2.1
	10	Temperature (C)	0.016 [0.000, 0.062]	0.000 [0.000, 0.001]	0.95	1.1
	100	Temperature (C)	0.016 [0.000, 0.062]	0.000 [0.000, 0.001]	0.95	1.1
	1000	Temperature (C)	0.016 [0.000, 0.062]	0.000 [0.000, 0.001]	0.95	1.1
	10	Topsoil bulk density (R)	0.016 [0.001, 0.063]	0.000 [0.000, 0.001]	0.98	1.2
	100	Topsoil bulk density (R)	0.016 [0.001, 0.063]	0.000 [0.000, 0.001]	0.98	1.2
	1000	Topsoil bulk density (R)	0.016 [0.001, 0.063]	0.000 [0.000, 0.002]	0.98	1.2
	10	Topsoil clay (R)	0.029 [0.002, 0.080]	0.012 [-0.003, 0.033]	0.97	1.2
	100	Topsoil clay (R)	0.034 [0.003, 0.090]	0.016 [-0.009, 0.045]	0.96	1.3
	1000	Topsoil clay (R)	0.034 [0.006, 0.092]	0.016 [-0.006, 0.051]	0.96	1.3
	10	Topsoil sand (C)	0.023 [0.001, 0.072]	0.005 [0.000, 0.019]	0.97	1.3
	100	Topsoil sand (C)	0.024 [0.001, 0.073]	0.007 [0.001, 0.025]	0.97	1.5

	1000	Topsoil sand (C)	0.025 [0.001, 0.073]	0.008 [0.001, 0.026]	0.97	1.5
	10	Topsoil silt (C)	0.024 [0.005, 0.073]	0.008 [-0.007, 0.022]	0.95	2.0
	100	Topsoil silt (C)	0.027 [0.007, 0.075]	0.010 [-0.012, 0.029]	0.94	2.9
	1000	Topsoil silt (C)	0.027 [0.005, 0.075]	0.010 [-0.013, 0.030]	0.94	3.0
	10	Urban areas (R)	0.018 [0.002, 0.072]	0.002 [0.000, 0.010]	1.00	1.1
	100	Urban areas (R)	0.054 [0.010, 0.148]	0.034 [-0.002, 0.116]	0.99	1.3
NSS	10	Cattle (R)	0.017 [0.001, 0.093]	0.004 [0.000, 0.014]	0.98	2.1
	10	Pastures (C)	0.023 [0.002, 0.106]	0.009 [0.001, 0.027]	0.99	2.1
	100	Pastures (C)	0.026 [0.002, 0.105]	0.012 [0.001, 0.038]	0.98	2.9
	1000	Pastures (C)	0.027 [0.002, 0.106]	0.012 [0.001, 0.039]	0.98	3.3
	10	Primary forest (C)	0.041 [0.006, 0.089]	0.005 [0.000, 0.012]	0.98	3.1
	100	Primary forest (C)	0.042 [0.008, 0.094]	0.007 [-0.001, 0.017]	0.97	4.2
	1000	Primary forest (C)	0.044 [0.009, 0.096]	0.008 [-0.001, 0.020]	0.97	4.4
	10	Primary non-forests (R)	0.035 [0.005, 0.118]	0.020 [0.000, 0.055]	0.98	14.0
	100	Primary non-forests (R)	0.052 [0.011, 0.136]	0.034 [-0.033, 0.099]	0.94	29.0
	1000	Primary non-forests (R)	0.054 [0.012, 0.138]	0.036 [-0.033, 0.111]	0.93	29.0
	10	Soil water storage capacity (C)	0.024 [0.002, 0.093]	0.009 [-0.001, 0.032]	0.94	1.4
	100	Soil water storage capacity (C)	0.029 [0.003, 0.087]	0.011 [-0.008, 0.038]	0.92	1.2
	1000	Soil water storage capacity (C)	0.029 [0.002, 0.087]	0.012 [-0.01, 0.039]	0.92	1.2
	10	Topsoil clay (R)	0.019 [0.001, 0.095]	0.006 [0.001, 0.017]	1.00	2.2
	100	Topsoil clay (R)	0.022 [0.001, 0.098]	0.008 [0.001, 0.023]	0.99	2.4
	1000	Topsoil clay (R)	0.023 [0.001, 0.100]	0.008 [0.001, 0.024]	0.99	2.4
	10	Topsoil sand (C)	0.015 [0.001, 0.088]	0.002 [0.000, 0.009]	0.99	2.2
	100	Topsoil sand (C)	0.016 [0.001, 0.091]	0.003 [0.000, 0.011]	0.99	2.3
	1000	Topsoil sand (C)	0.016 [0.001, 0.092]	0.003 [0.000, 0.011]	0.99	2.3

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